

Exploring Common Volatility in Cryptocurrencies: A Subsectoral Perspective

Son Duy Pham^a, Weihao Han^a, Hung Xuan Do^{b,d,1}, Linh Pham^c

^a *University of Aberdeen Business School, Dunbar Street, Aberdeen, United Kingdom*

^b *School of Accountancy, Economics and Finance, Massey University, New Zealand*

^c *Economics, Business, Finance Department, Lake Forest College, Lake Forest, IL, USA*

^d *International School, Vietnam National University, Hanoi, Viet Nam*

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¹ Corresponding author: School of Accountancy, Economics and Finance, Massey University, New Zealand (h.do@massey.ac.uk). Other authors' emails: Son D. Pham (son.pham@abdn.ac.uk), Weihao Han (weihao.han@abdn.ac.uk), Linh Pham (lpham@lakeforest.edu).

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Abstract

Cryptocurrencies offer the potential for high returns but are subject to extreme volatility driven by economic shifts, technological developments, and regulatory changes, as exemplified by events like the Terra collapse and FTX bankruptcy. Although all cryptocurrencies share fundamental features, such as decentralization and blockchain-based infrastructure, they may differ significantly in design, consensus mechanisms, and governance structures. These differences categorize them into distinct subsectors with unique risk profiles. This study applies the novel methodology proposed by Engle and Campos-Martins (2023) to analyse the common volatility (COVOL) across cryptocurrency subsectors, including Proof of Work (PoW), Proof of Stake (PoS), and Smart Contract platforms. Using a sample of the largest 17 cryptocurrencies from March 2016 to April 2024, our findings reveal distinct volatility patterns within these subsectors, driven by macroeconomic factors such as the term spread and interest rates. COVOL proves to be a strong predictor of market volatility and correlations, offering valuable insights into the dynamics of different cryptocurrency groups. The results provide essential guidance for investors and policymakers aiming to improve risk management and develop targeted strategies in the volatile cryptocurrency market.

JEL Classification Codes: C58, F31, G11, G15

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1. Introduction

The cryptocurrency market has experienced unprecedented growth over the past decade, evolving from a niche asset class to a significant component of global finance. As of November 2024, the market capitalisation of cryptocurrencies exceeded \$2.4 trillion,¹ driven by growing institutional interest, decentralised finance (DeFi) innovations, and the increasing adoption of blockchain technology across various sectors.² This remarkable expansion has not only attracted investors seeking high returns but has also garnered the attention of policymakers and researchers, who focused on understanding the risks inherent in such a dynamic and volatile market.

Although cryptocurrencies offer high return potential, the underlying extreme volatility, driven by macroeconomic shifts, technological advancements, regulatory actions, and geopolitical events are nontrivial. Events such as the Terra collapse, FTX bankruptcy, and central bank policy tightening have underscored these risks. While all cryptocurrencies share common characteristics such as decentralisation, reliance on cryptographic principles, and blockchain-based infrastructure, they differ substantially in their design features, consensus mechanisms, governance structures, and intended use cases – factors that fundamentally shape their risk profiles and market dynamics (Farell, 2018; Bolt and van Oordt, 2020; Milunovich, 2022; Wendl et al., 2023). For example, Eska et al. (2024) document significant differences in volatility across cryptocurrencies, with design features playing a key role. Older cryptocurrencies using Proof of Work (PoW) tend to be more stable, whereas newer Proof of Stake (PoS) cryptocurrencies are generally more volatile, along with being more sustainable and efficient due to lower energy consumption, reduced centralisation risk, and higher attack costs (Eska et al., 2024). Even though prior studies show these structural differences lead to varying volatility patterns

¹ See <https://coinmarketcap.com>.

² This rapid growth has attracted a wide range of investors, from retail to institutional, solidifying cryptocurrencies as a legitimate asset class. The number of cryptocurrencies has surged, with over 10,000 digital assets listed on CoinGecko, reflecting the industry's pace of innovation. New projects and decentralised applications are continually emerging, supporting diverse use cases such as decentralised finance (DeFi) and non-fungible tokens (NFTs). The DeFi market alone has a total value locked (TVL) exceeding \$50 billion, according to Binance Research, highlighting its potential. The rise of cryptocurrency exchange-traded funds (ETFs) and derivatives has further legitimised the market.

and distinct macroeconomic sensitivities between PoW and PoS assets, the latent common volatility measure that drives the co-movement of volatility across a class of assets remains undiscovered. Thus, our paper endeavours to bridge the gap by quantifying the common volatility of three cryptocurrency subsectors —namely, Proof of Work, Proof of Stake, and Smart Contract —and investigating the determinants and their mechanisms of common volatility. Analysing common volatility at the subsector level is crucial for understanding systemic risk, sector-specific vulnerabilities, and for developing targeted risk management strategies and regulatory frameworks.

We analyse the risk profiles of the subsectors of the cryptocurrency market via the Engle and Campos-Martins' (2023) common volatility (COVOL) measure. This is a novel approach that captures collective, time-varying volatility at the sector or market level. The rationale for adopting COVOL is twofold. *First*, unlike traditional volatility measures, which focus on individual asset price fluctuations, COVOL allows for assessing volatility that impacts an entire market or specific subsectors within it. This is particularly relevant for cryptocurrencies, where systemic events—such as regulatory crackdowns, macroeconomic disruptions, or technological failures—can trigger simultaneous price movements across various cryptocurrencies in the same subsector. As cryptocurrencies are highly interconnected, COVOL helps to capture these interlinked volatilities, providing a more comprehensive understanding of overall market risk.

Second, a key advantage of the COVOL measure is its ability to reflect time-varying nature, making it particularly useful in a market like cryptocurrency, where prices and volatility can change rapidly. Events such as exchange collapses, regulatory shifts, or macroeconomic disruptions can cause periods of heightened volatility, during which the market becomes more unstable. By applying COVOL, we assess how these shocks propagate across different market segments and identify which subsectors are more vulnerable during periods of stress. This is crucial not only for understanding volatility in the broad cryptocurrency market but also for examining how different subsectors react to shocks.

In our study, to facilitate interpretation, we regard each group of cryptocurrencies, categorised by different consensus mechanisms, as a sector. By this, we aim to contribute to the literature by exploring the COVOL measures across *distinct subsectors* of the cryptocurrency market, including PoW assets, PoS assets, and also extend to include other Smart Contract Platforms. These subsectors

serve different functions within the cryptocurrency ecosystem and are likely to exhibit varying levels of sensitivity to economic, regulatory, and technological shocks. For instance, as discussed earlier, PoW cryptocurrencies tend to be viewed as more established and stable, while PoS and Smart Contract Platforms are newer, more speculative, and highly innovative. Therefore, we aim to uncover the factors (and how) that drive common volatility in the subsectors of cryptocurrency, which may help to improve our understanding of varying levels of volatility across different cryptocurrencies.

Our study makes several key contributions to the literature on cryptocurrency markets. *First*, we are the first to quantify common volatility (COVOL) across different cryptocurrency subsectors using the novel methodology of Engle and Campos-Martins (2023). This method not only allows us capture the collective volatility response of cryptocurrencies to external shocks but also reveals the intricate volatility interactions within this rapidly evolving market. For instance, we note that the PoW subsector demonstrates high COVOL values during speculative bubbles, whereas COVOLs of the PoS subsectors increase amidst technological upgrades and regulatory changes, respectively. We assert that the hikes of COVOL of PoW coins mainly revolve around the price momentum because most PoW are large coins with steady hash rate and mining cost, which do not change overnight in a bubble. Conversely, PoS and Smart Contract peaks because network upgrades and regulatory announcements constitute systemic uncertainties that impact every token in the subsector simultaneously, they produce highly correlated volatility shocks captured by the common-volatility factor. Our work contributes to the expanding body of research on cryptocurrency risk exposures, building on the insights of studies by Wang (2022), Wang et al. (2022a, 2022b), and Lucey et al. (2022).

Second, we extend the asset pricing literature on the cryptocurrency market (e.g., Gregoriou, 2019; Dunbar and Owusu-Amoako, 2022; Han et al., 2024) by demonstrating the explanatory power of COVOL in predicting cryptocurrency risk and correlations. We find that subsector-wise COVOL can significantly explain the market volatility for PoW, PoS, and Smart Contract. Specifically, we note that a one-unit increase in COVOL is associated with 0.31, 0.15, and 0.09 increase in cryptocurrency market volatility (ϑ_m^{BMD}) for PoW, PoS, and Smart Contract, respectively. The positive relationship remains robust even after we change the proxy for cryptocurrency market volatility to subsection mean

correlation ($\rho_{subsector,m}$). This finding aligns with the work of Engle and Campos-Martins (2023) in global equity markets and underscores the importance of COVOL in understanding systemic risk within cryptocurrency subsectors. By showing that COVOL effectively captures the co-movement of cryptocurrency returns and their shared volatility, we offer a robust framework for evaluating risk in this highly volatile and interconnected market. This contribution is especially important for financial analysts and researchers, as it validates COVOL as a valuable tool for assessing the common risk factors in subsector cryptocurrency portfolios.

Third, we examine the drivers of common volatility within individual cryptocurrency subsectors, setting our study apart from prior research that either focuses on the determinants of volatility or returns for a single cryptocurrency (e.g., Wu et al., 2022; Wang et al., 2023a) or investigates the broader cryptocurrency market as a whole (e.g., Bouri et al., 2021b; Wang et al., 2023b). By analysing macroeconomic indicators and market-specific factors, including the Cryptocurrency Policy Uncertainty Index ($UCRY_{Policy}$) and Cryptocurrency Price Uncertainty Index ($UCRY_{Price}$), we provide a comprehensive view of what could influence market movements in these subsectors. Some factors exert a homogeneous impact on COVOL across all cryptocurrency sub-sectors, while others have more sector-specific effects. For instance, we find that increases in the $UCRY_{Price}$ consistently intensify the co-movement of cryptocurrencies within each sub-sector, indicating a broad-based sensitivity to overall crypto market sentiment. In contrast, monetary policy uncertainty, which captures market uncertainty related to central bank actions, significantly amplifies the common volatility factor only within the PoS (Proof-of-Stake) sub-sector, suggesting a differentiated sensitivity to macro-financial conditions. By decomposing subsector-specific volatility determinants, we shed light on how determinants vary across sub-sectors, allowing for more tailored risk management strategies. This contribution highlights the need to consider the heterogeneous latent factors within the cryptocurrency market, which is essential for both risk diversification and regulatory oversight..

In summary, our study offers critical insights for both investors and policymakers by highlighting the importance of understanding and managing common volatility in the cryptocurrency market, particularly given the differences across subsectors. Investors can use these findings to inform

asset allocation, portfolio construction, and risk management strategies, while policymakers may leverage the understanding of COVOL dynamics to design stabilising regulations that promote long-term market resilience. The remainder of the paper is organised as follows: Section 2 provides a brief review of the relevant literature. Section 3 introduces the modelling framework applied to estimate COVOL in the cryptocurrency market. Section 4 describes the data and discusses the estimation results for COVOL. Section 5 delves into COVOL's dynamics, highlighting its ability to elucidate both the volatility and correlations in the cryptocurrency market, as well as the key determinants behind them. Finally, Section 6 provides concluding remarks.

2. Literature review

The literature on cryptocurrency markets is generally divided into three main strands: (1) trading characteristics, such as returns, volatility, and liquidity; (2) market efficiency and investor behaviour; and (3) the relationship between cryptocurrencies and traditional markets, particularly in terms of hedging and diversification benefits.

The first strand focuses on the trading characteristics of cryptocurrencies, with particular attention to their returns, volatility, and liquidity. Cryptocurrencies are often found to provide higher average daily returns compared to traditional assets (Lee et al., 2018; Trimborn et al., 2020; Petukhina et al., 2021). Several studies examine the predictability of these returns, identifying determinants such as technical patterns (Bianchi et al., 2022), cross-cryptocurrency influences (Guo et al., 2024), trading volume (Bouri et al., 2019), and speculative behaviour (Koutmos and Payne, 2021). Despite their attractive returns, cryptocurrencies are marked by extreme volatility, driven by factors including trading volume, global uncertainties, Google search activity, and stock market performance (Bouri et al., 2021b; Wang et al., 2023b). Volatility spillovers among cryptocurrencies have also been widely documented (Yi et al., 2018; Ji et al., 2019a; Bouri et al., 2021a). Liquidity remains a key research focus, with studies exploring its role in cryptocurrency markets (Amihud, 2002; Brauneis et al., 2021; Zhang and Li, 2021; Bianchi et al., 2022). Unlike previous studies treating cryptocurrencies as individual assets, we gather cryptocurrencies based on their mechanisms in the blockchain and is the first to quantify a latent

common volatility at the subsector level. We reveal the heterogeneity in shared systematic risks and propose the new risk management framework.

The second strand investigates market efficiency and investor behaviour within cryptocurrency markets. Studies frequently report inefficiencies, evidenced by abnormal returns (Gregoriou, 2019) and directional predictability (Fousekis and Grigoriadis, 2021). Herding behaviour, often driven by social media, is another critical area of research (Kallinterakis and Wang, 2019; King and Koutmos, 2021). Social influence and financial literacy also shape trading behaviour (Gupta et al., 2020; Nadler and Guo, 2020). Additionally, market efficiency is shown to vary with the level of market development (Petukhina et al., 2021; Vidal-Tomás, 2021), emphasising the evolving nature of cryptocurrency markets compared to traditional financial assets. Our paper departs from the second strand of the literature that illuminates the return predictability and herding dynamics. We focus on systematic components of volatility across subsectors and estimate the loadings using a one-factor common volatility model for all subsectors, aiming to unearth the macroeconomic and crypto-specific drivers behind the volatility spikes.

The third strand explores the relationship between cryptocurrencies and traditional markets, particularly their connectedness, hedging capabilities, and diversification benefits. Research indicates weak connectedness between Bitcoin and traditional assets (Zeng et al., 2020), although some studies find significant spillover effects (Andrada-Félix et al., 2020). Cryptocurrencies, especially Bitcoin, are often viewed as effective hedging tools against stock market risks, outperforming traditional assets such as gold and commodities (Bouri et al., 2020a; Bouri et al., 2020b). This hedging ability has proven particularly robust during periods of market turmoil (Corbet et al., 2020; Koutmos et al., 2021; Mariana et al., 2021). Furthermore, cryptocurrencies provide diversification benefits when added to portfolios containing equities (Demiralay and Bayraci, 2021; Anyfantaki et al., 2021; Petukhina et al., 2021), energy commodities (Ji et al., 2019b; Okorie and Lin, 2020; Pham et al., 2022), and gold (Kumah et al., 2022). Recent findings suggest that diversifying across cryptocurrency subsectors can further enhance risk-adjusted returns (Huang et al., 2023). The previous studies focus on how cryptocurrencies interact with other asset classes such as stocks, commodities, precious metal for diversification benefits, while our paper sheds light on the internal dynamics of the crypto market from a risk management aspect. We

aim to propose a volatility-centred framework that not only complements the connectedness studies but also specify a new tool to hedge within the digital-asset universe.

To emphasise, despite these advancements, several research gaps persist in understanding the volatility dynamics of the cryptocurrency sector. A crucial unresolved question is whether a common volatility factor drives fluctuations across cryptocurrency subsectors. According to Farrell (2018), Bolt and van Oordt (2020), Wendl et al. (2023), cryptocurrency sub-sectors have varied fundamental characteristics, which ultimately affect their common volatility factor. Identifying such a factor is vital for assessing systemic risks and understanding how market events collectively impact digital assets. Moreover, it is important to analyse the drivers of this common volatility, including macroeconomic conditions, investor sentiment, regulatory developments, and global financial risks. Another key issue is whether different cryptocurrencies have varying exposures to this common volatility. This study aims to address these gaps by employing the novel common volatility measure developed by Engle and Campos-Martins (2023) to assess common shocks across cryptocurrency subsectors. By analysing COVOL, this research seeks to uncover how common shocks vary across subsectors and how they are influenced by macroeconomic variables and other financial risks.

3. Methodology: The COVOL measure

We estimate the common volatility for each cryptocurrency subsector using the procedure outlined in Engle and Campos-Martins (2023). First, let (r_t) be the vector of cryptocurrency excess daily returns $r_t = (r_{1,t}, \dots, r_{N,t})'$, where $(r_{i,t} = \tilde{r}_{i,t} - r_{f,t})$. Here, $\tilde{r}_{i,t}$ is the observed return, and $r_{f,t}$ is the risk-free return for $i = 1, \dots, N$ and N represents the number of cryptocurrencies in each subsector. The risk-free rate benchmark is the yield on the U.S. one-year Treasury notes. The daily risk-free rate is calculated by dividing the annualised risk-free rate by 360.

The next step in calculating common volatility involves a factor model with GARCH(1,1) errors for each series of excess returns $r_{i,t}$ as follows:

$$r_{i,t} = c_i + \delta_i r_{i,t-1} + \beta_i f_t + u_{i,t} \quad (1)$$

$$u_{i,t} = \sqrt{h_{i,t}} e_{i,t} \quad (2)$$

$$h_{i,t} = \omega_t + \alpha_{i,t}u_{i,t-1}^2 + \beta_{i,t}h_{i,t-1} \quad (3)$$

where i, t denotes a specific cryptocurrency and time. $c_i, \delta_i, \beta_i', \omega_t, \alpha_{i,t}, \beta_{i,t}$ are parameters to be estimated from the GARCH(1,1) model and we have ($|\delta_i| < 1; \omega_t > 0; \alpha_{i,t} > 0; \beta_{i,t} \geq 0; \alpha_{i,t} + \beta_{i,t} < 1$). f_t is a factor vector, including the first principal component of cryptocurrencies' excess return series for each subsector.

From the estimation of GARCH(1) model for cryptocurrencies' excess returns, we get the vector of standardised residuals $e_t = (e_{1,t}, \dots, e_{N,t})'$. According to Engle and Campos-Martins (2023), even though the standardised residuals have unit variance and zero covariance, their squared term or absolute values are likely to be correlated in the cross-section. Therefore, the comovement of volatilities is most likely caused by the positive correlation between shocks to those volatilities, since volatility is partly predictable. Assume that the variance shock to cryptocurrency i is:

$$\varphi_{i,t}^\sigma \equiv \frac{u_{i,t}^2 - h_{i,t}}{h_{i,t}} = e_{i,t}^2 - 1 \quad (4)$$

where $\varphi_{i,t}^\sigma$ is the proportional difference between the squared idiosyncrasy $e_{i,t}^2$ and its expectation 1. If many cryptocurrencies have larger squared idiosyncrasies than usual at the same time, this can be viewed as a common volatility shock to the entire cryptocurrency subsector. Let x_t^σ be the variance (latent) factor that captures the common volatility (COVOL) in the cryptocurrency subsector and $x_t^\sigma > 0; E[x_t^\sigma] = 1$.

To test whether the common volatility (x_t^σ) of the cryptocurrency subsector exists, we conduct the following test proposed in Engle and Campos-Martins (2023). Let ρ_{e^2} be the equicorrelation of the squared standardised residuals. The test-statistics for the existence of common volatility among the selected cryptocurrencies is given as follows:

$$T_{e^2} = \sqrt{\frac{NT}{(N-1)/2} \frac{\sum_{i>j=1}^N \sum_{t=1}^T (e_{i,t}^2 - 1)(e_{j,t}^2 - 1)}{\sum_{i=1}^N \sum_{t=1}^T (e_{i,t}^2 - 1)^2}} \quad (5)$$

T_{e^2} follows a normal distribution under the null hypothesis of no common volatility within the cryptocurrency subsector or $\rho_{e^2} = 0$. If we can reject the null hypothesis of the test shown in Eq. (5), we can further decompose the standardised residuals as follows:

$$e_{i,t} = \sqrt{g(s_i, x_t^\sigma)} \epsilon_{i,t} \quad (6)$$

$$g(s_i, x_t^\sigma) = s_i(x_t^\sigma - 1) + 1 \quad (7)$$

where $\epsilon_{i,t}$ is independently and identically normally distributed with zero mean and unit variance, $\epsilon_{i,t} \sim IIN(0,1)$ with $i = 1, \dots, N$. Moreover, vector $\epsilon_t = (\epsilon_{1,t}, \dots, \epsilon_{N,t})'$ is independent of x_t^σ . In Eq. (7), s_i is the factor loading for cryptocurrency i . The estimation of x_t^σ and s_i is conducted using the maximum likelihood method. x_t^σ indicates the time-varying common volatility or COVOL of the subsector and s_i represents the sensitivity of cryptocurrency i to COVOL of the subsector. The higher the value of s_i , the more susceptible cryptocurrency i is to systematic risk within the subsector market.

4. Data

To estimate COVOL, we collected daily price data for the 17 largest cryptocurrencies across three subsectors: Proof of Work (PoW), Proof of Stake (PoS), and Smart Contract Platform systems. Initially, we selected the 20 largest cryptocurrencies by market capitalisation as of April 10, 2024, using data from coingecko.com. We then categorised these cryptocurrencies into the three key subsectors and excluded three that could not be classified. Coingecko.com is a trusted source for cryptocurrency prices, trading volumes, and market capitalisation, aggregating information from over 400 major cryptocurrency exchanges (Han et al., 2023). Additionally, we excluded stablecoins from the sample because their values are pegged to fiat currencies, resulting in minimal price fluctuations.

The sample period spans from 11 March, 2016 to April 23, 2024, covering significant global financial and cryptocurrency market events. The starting date was chosen to ensure that the sample includes at least two cryptocurrencies in a subsector at any point in time. Since cryptocurrencies have different launch dates, the resulting dataset is an unbalanced panel, comprising 17 cryptocurrencies, each with at least one year of price history. Appendix A1 provides a detailed list of the cryptocurrencies

included in the sample, along with their respective tickers, descriptions, and market capitalisations. Notably, Bitcoin (BTC) and Ethereum (ETH) are the two largest cryptocurrencies, with market capitalisations far exceeding those of others in the sample.

We calculate the daily logarithmic return for each cryptocurrency from the daily price series.³ Next, we compute the excess returns ($r_{i,t}$) by subtracting the daily risk-free interest rate from the daily returns of each cryptocurrency. The U.S. 1-year Treasury note yields are used as a proxy for the risk-free rate (r_f). The data for U.S. 1-year Treasury note yields is sourced from the St. Louis Federal Reserve website.⁴

Table 1 presents the descriptive statistics of excess returns for cryptocurrencies across the Proof of Work (PoW) (Panel A), Proof of Stake (PoS) (Panel B), and Smart Contract Platform (Panel C) subsectors. As Ethereum (ETH) transitioned from proof-of-work (PoW) to proof-of-stake (PoS) on September 15, 2022, it is classified as part of the PoW sample for periods prior to this date and as part of the PoS sample for periods from this date onward.

Across all subsectors, most cryptocurrencies exhibit positive mean excess returns, highlighting their overall profitability during the sample period. In the PoW subsector, as presented in Panel A, Ethereum (ETH) and Bitcoin (BTC) demonstrate the highest mean excess returns, at 0.09% and 0.07%, respectively. Panel B shows that the PoS subsector is led by Solana (SOL) and Avalanche (AVAX), with mean excess returns of 0.15% and 0.08%, respectively. Meanwhile, Panel C highlights Binance Coin (BNB) as the leader in the Smart Contract Platform subsector, achieving a mean return of 0.07%. Among the selected cryptocurrencies, only ICP and FIL report negative mean excess returns over the research period.

The standard deviation values quantify the volatility, with PoW cryptocurrencies like Dogecoin (DOGE) and PoS coins like Matic (MATIC) having higher standard deviations, indicating greater fluctuations compared to more established assets like Bitcoin (BTC) and Ethereum (ETH). Across most

³ The daily logarithm return series of cryptocurrencies are plotted in Appendix A2.

⁴ See <https://fred.stlouisfed.org/series/DGS1>

PoW and PoS cryptocurrencies, skewness values indicate positive asymmetry in returns, particularly for Dogecoin (DOGE), which has the highest skewness of 4.71. By contrast, the majority of Smart Contract Platform cryptocurrencies display negative skewness. High kurtosis values for most cryptocurrencies reflect the presence of extreme price movements, with Dogecoin (DOGE) and Binance Coin (BNB) being particularly prone to price spikes.

The Jarque-Bera (JB) test results strongly reject the hypothesis of normality for all cryptocurrencies, confirming non-normal distributions across the board. Augmented Dickey-Fuller (ADF) tests show that returns are stationary for all cryptocurrencies. Significant autocorrelation, revealed by Ljung-Box Q and Q² tests, is present in both returns and squared returns for all cryptocurrencies in the sample.

[Please insert Table 1 about here]

5. Empirical results

5.1. Cryptocurrency subsector COVOL

5.1.1. Proof of Work (PoW) subsector

Prior to estimating the PoW subsector's common volatility, we test the null hypothesis of no common variance shocks, as detailed in Eq. (5). For PoW cryptocurrencies, the correlation of the squared standardised residuals is $\rho_{e^2, PoW} = 0.2832$ with the test statistic of $T_{e^2, PoW} = 41.31$ and a p -value of 0.00001. This strongly rejects the null hypothesis, indicating the existence of common volatility among the PoW cryptocurrencies. We establish and PoW COVOL via the methodology in Section 3, and visualise its 20-day moving average in Figure 1. Subsequently, Table 2 presents the dates with the most extreme COVOL values and the loading factors of each cryptocurrency.

Panel A of Table 2 shows the largest estimates of common volatility (COVOL) for the PoW subsector ($\chi_{PoW,t}^\sigma$).⁵ For comparative analysis, we also present the returns on the S&P 500 index (r_{SP500}

⁵ The 20-day moving average of the COVOL for the PoW subsector is shown in Appendix Online A3 Figure 1.

), WTI crude oil futures (r_{WTI}), and spot gold (r_{Gold}) corresponding to the dates of the highest cryptocurrency COVOL values along with the cross-sectional daily mean returns of PoW cryptocurrencies ($\bar{r}_{PoW}$). The extreme COVOL values provide important insights into the sensitivity of the cryptocurrency market, particularly PoW cryptocurrencies, to sector-specific and global financial events.

In Table 2, most largest COVOL values is associated with significantly positive sector returns. For instance, the highest COVOL value, 152.72, was recorded on January 28, 2021, when the mean return for PoW cryptocurrencies ($\bar{r}_{PoW}$) was 14.47%. This coincided with the cryptocurrency market bubble in 2021. This period was characterised by a rapid surge in cryptocurrencies' price, primarily driven by speculative investments, causing market-wide volatility to spike. Interestingly, the S&P 500 and WTI crude oil futures exhibited contrasting behaviour on this date, with modest gains in equities (+0.97%) and a drop in crude oil prices (-0.97%), while gold remained relatively stable (-0.20%). This suggests that during speculative bubbles, PoW cryptocurrencies can experience extreme volatility independent of general market trends.

On March 2, 2024, COVOL surged to 51.42 following record inflows into Bitcoin ETFs, illustrating how institutional investments can significantly amplify volatility in Proof of Work (PoW) cryptocurrencies. While Bitcoin ETFs provide more market access and liquidity, the influx of capital can trigger rapid price surges and sharp corrections, reflecting the cryptocurrency market's sensitivity to sector-specific developments. This event underscores the double-edged nature of institutional involvement, where increased legitimacy and growth, driven by ETF approval and investor demand, also heighten systemic risks by magnifying market volatility. The volatility caused by security-related incidents is also evident. For example, on October 11, 2018, the launch of Fidelity's institutional platform for cryptocurrencies triggered a surge in COVOL to 33.23.

The COVID-19 pandemic represents an important case where global market stress is aligned with heightened cryptocurrency volatility. On March 12, 2020, COVOL spiked to 25.85 amid a worldwide sell-off that saw the S&P 500 plummet by 10%, WTI fall by 4.59%, and gold decrease by 3.61%. This event illustrates how cryptocurrency volatility can align closely with broader financial

market movements under extreme global market conditions, deviating from the more common pattern of sector-specific volatility triggers. We assert that PoW coins are more vulnerable to internal systematic risks (e.g., cryptocurrency market bubble) compared to natural disaster (e.g., COVID-19) since the former COVOL values is almost 6 times larger than the later. This feature emphasises the safe haven property of PoW cryptocurrency, aligning with the findings of Bouri et al. (2020a). In addition to these events, institutional developments have repeatedly influenced COVOL values. For instance, the announcement of Bakkt, a federally regulated digital asset exchange by the NYSE on August 31, 2018, resulted in a COVOL value of 24.71, reflecting the impact of regulatory and infrastructure advancements on market volatility.

The factor loadings (s_i) of individual cryptocurrencies, shown in Table 2 Panel B, provide a granular view of their sensitivity to the sector-wide volatility. Dogecoin (DOGE) has the highest factor loading of 0.4587, indicating that it is particularly vulnerable to common volatility shocks, likely due to its speculative nature and high exposure to social media-driven trading activities. Similarly, Ethereum Classic (ETC) and Bitcoin Cash (BCH), with factor loadings of 0.4287 and 0.3994, respectively, also exhibit high sensitivity, potentially due to their derivative positioning relative to more established cryptocurrencies. In contrast, Bitcoin (BTC) and Ethereum (ETH), with a factor loading of 0.3773 and 0.3826, demonstrate relatively lower sensitivity, reflecting their more established status and broader investor base. Nonetheless, BTC and ETH's loading factors still indicate a significant exposure to common volatility, consistent with its role as the market leader.

The analysis of COVOL in the PoW subsector highlights how key events, both within the cryptocurrency sector and in global financial markets, can drive extreme volatility. Speculative bubbles, institutional developments, and security-related incidents all contribute to heightened market-wide volatility. The factor loadings reveal that while all PoW cryptocurrencies are sensitive to these volatility shocks, the extent of this sensitivity varies, with Dogecoin (DOGE) and Bitcoin Cash (BCH) being particularly exposed, while Bitcoin (BTC), Ethereum (ETH), and Litecoin (LTC) exhibit comparatively lower volatility sensitivity. For investors, this underscores the need for vigilant risk management when investing in PoW cryptocurrencies, particularly during periods of market turbulence.

[Please insert Table 2 in here]

5.1.2. Proof of Stake (PoS) subsector

Table 3 presents the most extreme COVOL values and individual cryptocurrencies' factor loadings in the PoS subsector.⁶ For PoS cryptocurrencies, the correlation of the squared standardised residuals is $\rho_{e^2, PoS} = 0.1123$ with the test statistic of $T_{e^2, PoS} = 17.26$ and a p -value of 0.00001. These results strongly reject the null hypothesis, providing evidence of common volatility among PoS cryptocurrencies. Table 3 provides insights into the largest estimated common volatility (COVOL) values for the Proof-of-Stake (PoS) subsector of cryptocurrencies. Similar to the PoW subsector, PoS cryptocurrencies experience significant volatility spikes, driven by a mix of sector-specific developments and economic factors. The most significant COVOL values in the PoS subsector reflect major market disruptions, regulatory changes, and technological upgrades.

The largest spike in COVOL occurred on December 10, 2019, with a value of 45.43, triggered by the closure of Houbi's US-based crypto trading platform. The shutdown of a major trading platform introduced uncertainty into the market, causing price tumbling across PoS cryptocurrencies, with the sector mean return ($\bar{r}_{PoS}$) of -15.40%. Despite this being a sector-specific event, it reflects how regulatory actions or business decisions can have a profound impact on the PoS market, highlighting the market's sensitivity to infrastructure changes within the cryptocurrency ecosystem.

Another significant spike in COVOL was observed on October 15, 2021, reaching a value of 29.13. This date aligns with Trustology, a prominent crypto custody firm, receiving full FCA registration as a UK-registered cryptoassets business, a milestone that bolstered confidence in the regulatory legitimacy of the crypto sector. Additionally, this period marked the height of a crypto bull market, fuelled by unprecedented monetary easing and globally low interest rates implemented to mitigate the economic fallout from the COVID-19 pandemic. The combination of increased regulatory

⁶ The 20-day moving average of the COVOL for the PoS subsector is shown in Appendix Online A3 Figure 2.

clarity and favourable macroeconomic conditions contributed to heightened investor optimism and trading activity, amplifying common volatility within the cryptocurrency market.

A crucial spike in COVOL was observed on July 30, 2022, in the aftermath of the cryptocurrency market bubble burst. This period followed the catastrophic collapse of the stablecoin TerraUSD in May 2022, which sent shockwaves across the broader crypto ecosystem, eroding investor confidence. Coupled with rising inflation and tightening monetary policies globally, the market experienced significant sell-offs, leading to a dramatic freefall in the valuations of many PoS cryptocurrencies.⁷ This heightened volatility reflected the fragility of the market during this turbulent period, as investors grappled with the compounded effects of systemic failures and adverse macroeconomic conditions. Likewise, in August 2019, a spike in COVOL (27.48) arose from U.S. regulatory developments, particularly a bill from the House of Representatives to eliminate double taxation on crypto transactions. Regulatory clarity often fuels volatility in cryptocurrency markets as investors respond to potential legal changes. Here, the proposed legislation benefited the crypto industry and led to increased market activity and volatility.

Another notable COVOL spike occurred in August 2021 (19.97), following the Ethereum London Hard Fork upgrade (EIP-1559). This upgrade introduced significant changes to Ethereum's fee structure, creating uncertainty in the market as investors and developers adjusted to the new system. This volatility spike reflects how technological upgrades and network changes within PoS cryptocurrencies can drive sharp price movements, especially for assets like Ethereum, which are integral to the broader cryptocurrency ecosystem.

In addition to these findings, it is noteworthy that several spikes in the COVOL of PoS cryptocurrencies were observed during the COVID-19 pandemic period. For instance, significant spikes in May and July 2020 coincided with a period of heightened volatility in global financial markets. Initially, this volatility was driven by widespread fear and uncertainty as the pandemic unfolded, leading

⁷ For instance, the price of Cardano (ADA) fell from 3 USD in October 2021 to around 0.4 USD in July 2022.

to a sharp sell-off in risky assets. However, this was soon followed by a phase of renewed investor appetite for risk, fuelled by unprecedented monetary easing and fiscal stimulus measures implemented worldwide to stabilise economies. These actions not only supported traditional financial markets but also spilled over into the cryptocurrency market, amplifying price fluctuations and contributing to the observed spikes in COVOL for PoS cryptocurrencies.

Panel B of Table 3 provides factor loadings (s_i) for individual PoS cryptocurrencies. A higher factor loading means the asset is more prone to sector-specific shocks. Cardano (ADA) has the highest factor loading (0.3972), suggesting that it is highly susceptible to market volatility as a newer and less established cryptocurrency. Aptos (APT) follows closely at 0.3892, indicating that despite its widespread use, it remains vulnerable to speculative shifts in the market, likely due to its role as a leading PoS platform. Polygon (MATIC) also shows a high factor loading (0.3791), given its pivotal role in Ethereum scaling and the DeFi ecosystem, making it sensitive to market dynamics. Ethereum (ETH) has the lowest factor loading of 0.2698, reflecting its greater market maturity and relative stability compared to smaller PoS assets. Overall, while all PoS cryptocurrencies are impacted by sector-wide volatility, larger and more established assets like Ethereum, Solana, and Avalanche exhibit relatively greater resilience compared to newer or smaller cryptocurrencies.

[Please insert Table 3 in here]

5.1.3. *Smart contract platform subsector*

For smart contract platform cryptocurrencies, the correlation of the squared standardised residuals is $\rho_{e^2, Smart} = 0.3015$ with the test statistic $T_{e^2, PoS} = 26.45$ and a p -value of 0.00001. These findings strongly reject the null hypothesis, confirming the presence of common volatility among smart contract platform cryptocurrencies. Table 4 presents the most extreme COVOL values and individual cryptocurrencies' factor loadings in the smart contract platform subsector, respectively.⁸

⁸ The 20-day moving average of the COVOL for the Smart Contract Platform subsector is shown in Appendix Online A3 Figure 3.

Panel A of Table 4 presents the largest estimated common volatility (COVOL) for the Smart Contract Platform subsector, highlighting dates of heightened volatility and the corresponding market events that drove these extreme values. The highest COVOL value of 49.26 was recorded on March 31, 2021, during the 2021 cryptocurrency market bubble, when the sector mean return ($\bar{r}_{Smart}$) was 5.26%. This period saw a massive surge in speculative investments across the cryptocurrency space, including smart contract platforms such as Ethereum, which experienced rapid price increases. The S&P 500 saw moderate gains (+0.36%), while WTI crude oil prices dropped (-2.32%) and gold rose (+1.31%), indicating broader market movements, but the extreme COVOL value highlights how the speculative frenzy in the cryptocurrency sector outpaced market volatility. This event reflects the overvaluation that characterised the 2021 bubble, which led to sharp corrections later. Similar findings are noted for the PoW and PoS subsectors as well, indicating the effectiveness of using COVOL to capture systematic risks.

On April 21, 2022, COVOL reached 29.03 during the cryptocurrency market bubble burst. This event marked a sharp correction following the speculative highs of 2021, with many cryptocurrencies experiencing significant losses in value. Another significant spike in COVOL occurred on September 3, 2020 (27.23), during a market correction. The S&P 500 suffered a sharp drop (-3.57%), which mirrors the volatility in the smart contract platforms. This period followed a strong rally in the crypto market earlier in the year, and the correction reflected a broader risk-off sentiment in global markets as investors reassessed their exposure to risky assets, including cryptocurrencies.

On September 24, 2019, COVOL reached 24.45, driven by a sell-off in the cryptocurrency market, particularly in Bitcoin, which dropped below USD 10,000. The S&P 500 and WTI crude oil both fell (-0.85% and -2.33%, respectively), while gold increased (+0.67%), reflecting a risk-averse environment where investors sought safe-haven assets like gold. The decline in cryptocurrency prices during this period, including that of smart contract platforms, was linked to uncertainty surrounding regulatory developments and market corrections following speculative rallies. On November 28, 2018, COVOL spiked to 21.09 when Bitcoin's market cap fell below USD 100 billion for the first time since October 2017. The sharp decline in Bitcoin's price, alongside widespread selloffs in other smart contract platforms, led to significant volatility.

The cryptocurrency market bubble in 2023 led to another notable spike in COVOL of smart contract platform cryptocurrencies, reaching 20.21 on October 21, 2023. This surge was driven by easing inflation in developed countries, fuelling speculation that central banks would reduce benchmark interest rates, thereby increasing demand for risky assets, including cryptocurrencies. The year also marked significant growth in the Ethereum ecosystem, with the number of Ethereum and wallet SDK installs hitting all-time highs for the fifth consecutive year, increasing by 31% and 126% year-over-year, respectively. Furthermore, the creation of EVM smart contracts across Ethereum Mainnet, Arbitrum, Optimism, and Polygon skyrocketed by 303% year-over-year in 2023, reflecting the rapid expansion and adoption of blockchain technologies during this period.⁹

[Please insert Table 4 in here]

Panel B of Table 4 shows the factor loadings for smart contract platform cryptocurrencies, highlighting their sensitivity to sector volatility. The listed cryptocurrencies—TRON (TRX), Internet Computer (ICP), Binance Coin (BNB), and Chainlink (LINK)—exhibit varying sensitivity levels. TRON (TRX) has the highest factor loading at 0.5136, indicating vulnerability to common shocks in the subsector. As an early smart contract platform for decentralised applications (dApps), TRON's price is driven by market sentiment, making it sensitive to cryptocurrency trends. Internet Computer (ICP) follows with a factor loading of 0.5056, reflecting exposure to volatility due to its decentralisation goals. Despite its potential, ICP's newness makes it prone to speculative fluctuations, causing strong price reactions to market sentiment changes. Binance Coin (BNB) has a factor loading of 0.4918 and remains sensitive to volatility; as the native token of the Binance exchange, its value is linked to the cryptocurrency ecosystem's health, especially during large market movements. Chainlink (LINK), with a factor loading of 0.4885, is also affected by smart contract subsector volatility. As a decentralised oracle network connecting smart contracts to real-world data, Chainlink's price is influenced by cryptocurrency market trends, especially during innovation or downturns.

⁹ Please see: <https://www.alchemy.com/blog/web3-developer-report-2023>

In summary, the analysis of sub-sector COVOL reveals that each segment of the cryptocurrency market—Proof-of-Work (PoW), Proof-of-Stake (PoS), and Smart Contract Platforms—exhibits distinct patterns of common volatility. This is evidenced by the differing dates on which each sub-sector experiences extreme COVOL episodes. Furthermore, the pairwise correlation coefficients among the three sub-sectors are relatively low, suggesting limited co-movement in their common volatility dynamics. Specifically, the correlations are 0.047 between PoW and PoS, 0.042 between PoW and Smart Contract Platforms, and 0.139 between PoS and Smart Contract Platforms.¹⁰ These findings underscore the heterogeneous nature of volatility transmission within the broader cryptocurrency market.

5.2. Validation tests

In the next two subsections, following Engle and Campos-Martins (2023), we conduct two validation tests to assess the validity of our COVOL measure.

5.2.1 Test 1: Predictive power of COVOL on cryptocurrency market volatility

First, we examine whether our subsector COVOL measures help explain the volatility of the overall cryptocurrency market. We use the S&P Cryptocurrency Broad Digital Market Index (BMD) as a proxy for the market. The BMD is one of the most comprehensive indices, capturing the fluctuations of 276 digital assets, including all cryptocurrencies in our sample.¹¹ In line with Engle and Campos-Martins (2023), we take the BMD squared standardised residuals (average over the calendar month) derived from the GARCH(1,1) model and denote them by $\vartheta_m^{BMD} = (e_m^{BMD})^2 - 1$. We then regress this realised cryptocurrency market volatility measure on different risk measures including the i) realised common volatility for each subsector ($COVOL_{subsector}^2$), ii) the monthly change in the Global Economic Policy Uncertainty Index (Δ_m^{GEPU}), iii) the monthly change in the implied volatility of the US stock market measured by VIX (Δ_m^{VIX}), and iv) the monthly change in the global geopolitical risk index by

¹⁰ To save space, we do not report these correlation coefficients in a table.

¹¹ The index was initiated by S&P Global on February 28, 2017.

Caldara and Iacoviello (2022) (Δ_m^{GOPRX}). As the common volatility ($COVOL_{subsector}^2$) indicates common shocks to each subsector, we expect it to be positively affect the volatility of the broad cryptocurrency market, proxied by ϑ_m^{BMD} .

The regression results of the first validation test are shown in Table 5 Panel A, with columns (1), (2), and (3) showing the results for the Proof of Work (PoW), Proof of Stake (PoS), and smart contract platforms (Smart) subsectors, respectively. Column 1 shows that the coefficient of the squared PoW common volatility measure is 0.31 with a standard error of 0.09, significant at the 1% level. This strong positive relationship indicates that increases in common volatility within the PoW subsector contribute substantially to the overall cryptocurrency market volatility. Given that PoW includes major cryptocurrencies like Bitcoin, which have significant market dominance, fluctuations in this subsector can have pronounced effects on the entire cryptocurrency market.

Column 2 shows that the coefficient on the PoS squared COVOL is 0.15 with a standard error of 0.08, significant at the 5% level. This positive relationship suggests that common volatility within PoS cryptocurrencies also impacts the broad volatility of the cryptocurrency market, though to a lesser extent than PoW. The PoS subsector includes assets like Ethereum after its transition to PoS, highlighting how shifts in this subsector's volatility can influence overall market dynamics. The Smart Contract platforms subsector squared COVOL (Column 3) shows a coefficient of 0.09 with a standard error of 0.04, significant at the 5% level. This result indicates that common volatility among smart contract platforms contributes positively to the overall market volatility. While the impact is smaller compared to PoW and PoS, it is still statistically significant, underscoring the role of this subsector in the broad market context.

Regarding the control variables, the monthly change in the Global Economic Policy Uncertainty Index (Δ_m^{GEPU}) does not exhibit a statistically significant impact across any subsector. This suggests that global economic policy uncertainties may not directly influence cryptocurrency market volatility, once subsector common volatility has been accounted for. The monthly change in the implied volatility of the US stock market (Δ_m^{VIX}) shows a positive and significant effect for the PoW and PoS subsectors, with coefficients of 0.17 and 0.21, respectively. This indicates that increasing volatility in

the US stock market can spill over into the cryptocurrency market through the PoW and PoS subsectors. However, Δ_m^{VIX} does not significantly affect the Smart Contract subsector. The monthly change in the Global Geopolitical Risk Index (Δ_m^{GPRX}) does not have a statistically significant effect in any of the models, implying that geopolitical risks might not have an immediate impact on cryptocurrency market volatility when subsector common volatility is considered.

In summary, the regression analysis in Table 5 Panel A demonstrates that the common volatility measures of the PoW, PoS, and Smart Contract subsectors play a significant role in explaining the volatility of the cryptocurrency market. The positive and statistically significant coefficients across all subsectors confirm that common shocks within these groups contribute to overall market volatility. These findings validate the role of common volatility (COVOL) factor in shaping subsector dynamics and underscore the need for both investors and policymakers to account for its effect when making decisions in the rapidly evolving cryptocurrency landscape.

5.2.2 Test 2: Predictive power of COVOL on the average correlation of cryptocurrency returns

In the second validation test, we investigate whether our common volatility measure can explain the average correlation of cryptocurrency returns within each subsector. Using the daily return series from our sample, we estimate all monthly pairwise correlation coefficients within each subsector and compute their monthly averages, denoted as $\rho_{subsector,m}$, where m denotes the calendar month. Since periods of high volatility typically coincide with higher correlations among assets, we expect increased correlations when the subsector's COVOL is elevated. To test this hypothesis, we regress $\rho_{subsector,m}$ on the subsector's COVOL and other measures of global uncertainties, including a one-period lagged variable $\rho_{subsector,m-1}$ to account for the persistence of the correlations.

The results of the second validation test, shown in Table 5 Panel B, highlight the impact of subsector common volatility (COVOL) on the average correlations of cryptocurrency returns within different subsectors. A 1-unit increase in the PoW squared COVOL is associated with a 2% increase in average correlation, emphasizing that periods of rising volatility make the movements of PoW cryptocurrencies more in sync with one another. Similarly, when the PoS squared COVOL increases by 1 unit, a 1.4% increase in correlations is observed, with a strong relationship between common

volatility and return synchronisation. The Smart Contract subsector exhibits a similar trend, though the magnitude is smaller. Across all subsectors, these findings confirm that higher COVOL corresponds with stronger co-movement among cryptocurrency returns.

The persistence of correlations over time is also noteworthy. The one-period lagged correlation term is highly significant across all subsectors, indicating that past correlation patterns have a strong influence on future movements. This persistence means that once cryptocurrencies within a subsector become more correlated, they are likely to remain that way for some time. Interestingly, global uncertainty factors, such as changes in economic policy (Δ_m^{GEPU}), U.S. stock market volatility (Δ_m^{VIX}), and geopolitical risk (Δ_m^{GOPRX}), do not significantly impact correlations within the subsectors. This suggests that the internal volatility within the cryptocurrency market is a more dominant driver of return correlations than external global risks.

For investors, this means that higher common volatility weakens the diversification benefits within subsectors, making it harder to reduce risk when correlations rise. Monitoring COVOL is essential for anticipating subsector movements. Policymakers, on the other hand, should recognise that systemic risks in the cryptocurrency market are driven more by market-specific volatility rather than economic or geopolitical uncertainties. This calls for a more focused regulatory approach that addresses internal market dynamics without unnecessarily hindering innovation. Overall, the results in Table 5 Panel B demonstrate how COVOL plays a crucial role in shaping the behaviour of cryptocurrencies, offering valuable insights for both market participants and regulators.

[Please insert Table 5 in here]

5.3. *Determinants of subsector COVOL*

5.3.1. *Models with daily data*

Given the significant variations and volatility observed in the COVOL measures, it is crucial for investors and policymakers to understand their key drivers. To uncover the factors influencing the COVOL measures, we employ both daily and monthly specifications. Our daily specification is specified by the equation below:

$$COVOL_t = \alpha + \beta_1 RAI_t + \beta_2 SP500_t + \beta_3 DXY_t + \beta_4 Term10Y2Y_t + \beta_5 DGS2_t + \varepsilon_t \quad (8)$$

where $COVOL_t$ is the daily COVOL measure for subsectors; α denotes the intercept; ε_t represents the error term; and β_1 to β_5 are estimated coefficients. The explanatory variables identified in Eq. (8) include i) the daily measure of risk aversion index (RAI) as a sentiment indicator (Bekaert et al., 2022); ii) the return on the S&P 500 index ($SP500$); iii) the US Dollar index (DXY); iv) the term spread as the difference between the yields of the U.S. 10-Treasury notes and that of the 2-year Treasury notes ($Term10Y2Y$); and v) the yields of the U.S. 2-year Treasury notes ($DGS2$).

These explanatory variables are chosen based on previous studies indicating their potential effects on cryptocurrency market returns, volatility, and overall common volatility. The RAI global investor sentiment index (Bekaert et al., 2022) notably affects cryptocurrency market dynamics (Huynh and Phan, 2023). Additionally, Wang et al. (2023a) propose that changes in the S&P 500 and US Dollar indexes serve as valuable indicators of Bitcoin's volatility. Moreover, monetary policy decisions can influence cryptocurrency volatility by impacting interest rates, liquidity, and broader economic conditions. Actions by central banks frequently alter investor risk tolerance and speculative behaviour (Claeys et al., 2018; Nguyen et al., 2019; Elsayed and Sousa, 2022; Che et al., 2023). For instance, during periods of expansionary monetary policy, such as lower interest rates or quantitative easing, the increased money supply generally leads to reduced returns on traditional assets like bonds. Consequently, investors may pivot towards riskier assets like cryptocurrencies, which increases their volatility.

The regression results of Eq. (8) for each subsector are presented in Table 6, highlighting the unique and varying influences of macroeconomic factors and market sentiment across different cryptocurrency subsectors: Proof of Work (PoW), Proof of Stake (PoS), and Smart Contract Platforms. For PoW cryptocurrencies, as displayed in the first column, the coefficient for RAI (investor sentiment) is negative but very small and statistically insignificant (-0.005). This suggests that market sentiment plays a minimal role in influencing volatility in this subsector, especially when compared to the broader cryptocurrency market. PoW assets like Bitcoin tend to be viewed as alternative stores of value and may be less reactive to short-term shifts in sentiment, instead drawing value from their established

position and adoption in the market. Similarly, the return on the S&P 500 (*SP500*) has a small positive but insignificant coefficient (0.0002), indicating that equity market performance has little to no influence on PoW volatility. This underscores the idea that PoW cryptocurrencies are somewhat decoupled from traditional equity markets, possibly because they are viewed as hedges against mainstream financial assets.

The US Dollar Index (*DXY*) shows a negative but insignificant relationship (-0.002) with PoW volatility, implying that fluctuations in the dollar's value have no meaningful impact on PoW assets. More importantly, the term spread (*Term10Y2Y*) exhibits a significant negative relationship (0.12), suggesting that a steeper yield curve, which typically signals economic growth and optimism, is associated with higher volatility in PoW cryptocurrencies. This result suggests that during periods of economic optimism, Proof-of-Work (PoW) cryptocurrencies, such as Bitcoin, attract speculative capital flows, resulting in increased trading activity and higher volatility. Alternatively, it could reflect a shift in perception where Bitcoin and other PoW assets are seen as speculative opportunities rather than safe-haven assets, amplifying their sensitivity to market conditions during such periods. Finally, the 2-year Treasury yield (*DGS2*) shows a negative but insignificant coefficient (-0.03), suggesting that short-term interest rate fluctuations have little effect on PoW volatility, further supporting the notion that this subsector responds differently to macroeconomic changes compared to the broader market.

For PoS cryptocurrencies, as seen in the second column, the coefficient for *RAI* is positive (0.32), though still statistically insignificant, indicating that market sentiment could have some influence on volatility in PoS cryptocurrencies, but the effect remains weak during the sample period. The coefficient for *SP500* is negative (-0.06) but insignificant, implying a weak and statistically insignificant inverse relationship between equity market performance and PoS volatility. This suggests that when equity markets perform poorly, investors may seek higher returns in PoS cryptocurrencies, potentially driving up volatility, though the effect is not strong enough to be conclusive.

The *DXY* coefficient remains negative (-0.005) and insignificant, reflecting the general finding that the value of the US dollar has little impact on cryptocurrency volatility, regardless of the subsector. However, the term spread (*Term10Y2Y*) is positive and significant (0.72), indicating that a steeper

yield curve increases volatility in PoS cryptocurrencies. This aligns with economic theory, as PoS assets, which are generally more speculative, tend to attract greater interest during periods of economic optimism and risk appetite, leading to heightened volatility. Similarly, the *DGS2* coefficient is positive and significant (0.29), suggesting that higher short-term interest rates lead to greater volatility in the PoS subsector. This result underscores the heightened sensitivity of PoS cryptocurrencies to shifts in monetary policy. In an environment of rising interest rates and tighter monetary conditions, investors are likely to reallocate capital away from these speculative assets, which often lack intrinsic value or stable cash flows. This capital shift can amplify market reactions, resulting in increased volatility for PoS assets. Additionally, the speculative nature of these assets may make them particularly vulnerable to changing investor risk appetites, further driving their sensitivity to monetary policy changes.

The third column focuses on Smart Contract Platforms, revealing unique dynamics in their response to market sentiment. The *RAI* coefficient, which measures the influence of risk aversion sentiment, is negative and statistically significant (-0.02). This finding suggests that as investor risk aversion increases, volatility within this subsector tends to decrease. This relationship may reflect the perception of smart contract platforms as more stable or utility-driven investments compared to other digital assets. Platforms like BNB (Binance Smart Chain) or similar technologies underpin significant blockchain activity, such as decentralised finance (DeFi) and non-fungible token (NFT) transactions. During periods of heightened risk aversion, these platforms may experience sustained utility-driven demand, reducing speculative trading and price swings. Moreover, the robust infrastructure and wide adoption of these platforms can provide a sense of stability for investors seeking exposure to blockchain technologies without taking on the extreme volatility associated with less established or highly speculative assets. As a result, the higher risk aversion sentiment acts as a stabilising factor, curbing excessive market fluctuations and leading to lower volatility in this subsector.

In addition, the *SP500* coefficient is positive but insignificant (0.02), indicating that equity market movements have a negligible influence on smart contract platforms' volatility. Similarly, the *DX1* coefficient remains negative and insignificant (-0.002), showing that currency fluctuations are not major drivers of volatility in this subsector. Furthermore, for Smart Contract Platforms, both the term spread (*Term10Y2Y*) and *DGS2* coefficients are positive but insignificant (0.01 for both), implying

that macroeconomic indicators like the yield curve and short-term interest rates have little impact on volatility in this subsector. This contrasts with the significant effects seen in the PoS subsector. The lack of impact from these macroeconomic variables suggests that smart contract platforms may be more insulated from traditional economic fluctuations, possibly due to their integral role in the emerging DeFi ecosystem, where technological advancements and adoption drive market dynamics more than economic indicators.

By comparing subsectors, we summarise stylised facts that reveal distinct sensitivities of cryptocurrency groups to market and economic factors. The PoS subsector shows greater responsiveness to macroeconomic conditions, with significant effects of term spread and short-term interest rates on volatility, reflecting their speculative nature and reliance on investor risk appetite. In the same vein, PoW cryptocurrencies demonstrate a positive relationship between term spread and volatility, indicating reduced volatility during economic optimism, likely due to their perceived stability and maturity. Meanwhile, smart contract platforms exhibit unique sensitivity to investor sentiment, with a negative *RAI* coefficient showing that increased risk aversion lowers their volatility. This suggests their utility-driven demand provides stability in cautious markets. These differences highlight how each subsector's characteristics shape its response to economic and market forces.

On one hand, our findings emphasise the complex interplay between macroeconomic factors and market sentiment in driving cryptocurrency volatility. Each subsector exhibits unique responses to these variables, underscoring the need for tailored investment strategies and regulatory approaches. On the other hand, understanding these distinct dynamics is essential for investors and policymakers aiming to effectively navigate the rapidly evolving and often volatile cryptocurrency market.¹²

[Please insert Table 6 in here]

¹² Further empirical results on the determinants of COVOL across sub-sectors during major recent crisis periods- specifically the COVID-19 pandemic and the Russia-Ukraine war- are provided in Online Appendix 4. These findings reinforce our overarching conclusion regarding the heterogeneous drivers of common volatility across different segments of the cryptocurrency market.

5.3.2. Monthly data models

Next, we convert all variables in section 5.3.2. to monthly frequencies and re-estimate the regression model in equation (8). This allows us to capture long-run trends among cryptocurrency subsector COVOLs and their determinants. The use of monthly data also allows us to include three additional explanatory variables:¹³ the U.S. monetary policy uncertainty (MPU) index developed by Baker et al. (2016), and two indices specific to the cryptocurrency market—cryptocurrency policy uncertainty ($UCRY_{Policy}$) and cryptocurrency price uncertainty ($UCRY_{Price}$), both text-based measures developed by Lucey et al. (2022) using LexisNexis Business data. This expanded model captures broader market conditions and the unique uncertainties within the cryptocurrency ecosystem.

The regression results of this enhanced model, presented in Table 7, show distinctive relationships across cryptocurrency subsectors. For Proof of Work (PoW) cryptocurrencies (Column (1)), the term spread ($Term10Y2Y$) continues to positively and significantly correlate with the common volatility with a coefficient of 0.11. This is consistent with the result using daily data. In addition, the $UCRY_{Policy}$ index coefficient is negative and significant (-0.012), indicating that higher cryptocurrency policy uncertainty decreases volatility in this subsector. This result suggests that PoW cryptocurrencies, such as Bitcoin, are less sensitive to regulatory and policy changes than other subsectors, possibly because they are more established and seen as "digital gold," making them more resilient to short-term regulatory changes. The fact that policy uncertainty lowers volatility in PoW assets could reflect market participants' perception that PoW cryptocurrencies can weather regulatory adjustments without major disruptions. By contrast, the estimated coefficient of $UCRY_{Price}$ is positive (0.002) and statistically significant at 5%, indicating that increased price uncertainty amplifies the common risk factor in PoW cryptocurrencies. This highlights the dual effect of policy and price dynamics on volatility within the PoW subsector.

¹³ These variables are only available at the monthly frequencies.

In the Proof of Stake (PoS) subsector (Column (2)), the results reveal a positive and significant relationship between the term spread ($Term10Y2Y$) and COVOL (0.55). This indicates that economic optimism, as reflected by a steeper yield curve, increases volatility in PoS cryptocurrencies, likely because these assets are perceived as higher-risk investments that attract more speculative capital in optimistic economic conditions. Interestingly, U.S. monetary policy uncertainty (MPU) has a positive and significant coefficient (0.91), suggesting that higher monetary policy uncertainty heightened common risk in PoS cryptocurrencies. This result may indicate a flight-to-safety behaviour where, during periods of high uncertainty about U.S. monetary policy, investors move assets into safer, less volatile investments, leading to increased volatility in PoS assets. Moreover, similar to the results for PoW cryptocurrencies, common volatility in the PoS subsector is positively and significantly related to price uncertainty factor, as demonstrated by the coefficient of $UCRY_{Price}$ (0.021). Other explanatory variables for PoS cryptocurrencies, including $UCRY_{Policy}$, are not statistically significant, indicating that these specific uncertainties may not play a major role in driving volatility within this subsector.

For Smart Contract Platforms (Column (3)), the results are more varied, reflecting the subsector's complexity and its integration into decentralised finance (DeFi) and decentralised applications (dApps). The S&P 500 index return shows a significant negative relationship with COVOL (-0.12), indicating that better equity market performance reduces volatility in this subsector. This result aligns with the view that when equity markets are performing well, investors might shift away from speculative assets like cryptocurrencies, thus reducing volatility in smart contract platforms. Conversely, both the $Term10Y2Y$ and $DGS2$ are positive and significant (0.09 and 0.05, respectively), suggesting that economic optimism and rising short-term interest rates increase volatility. Economic optimism likely spurs increased speculative activity on smart contract platforms, driving volatility, while higher short-term rates may increase volatility as these platforms attract more speculative capital chasing higher returns.

Additionally, $UCRY_{Policy}$ and $UCRY_{Price}$ exert significant impact on the smart contract platform subsector's COVOL. $UCRY_{Policy}$ has a negative coefficient (-0.01), implying that policy uncertainty reduces volatility in smart contract platforms. This could be because periods of high regulatory scrutiny

or uncertainty might cause investors to adopt a more cautious approach, limiting speculative trading and, thus, volatility. On the other hand, $UCRY_{Price}$ is positive and significant (0.004), suggesting that price uncertainty increases common volatility. This result underscores the inherently speculative nature of smart contract platforms, where fluctuations in price lead to greater instability as market participants respond more aggressively to price movements.

[Please insert Table 7 about here]

6. Conclusion

This paper provides valuable insights into the cryptocurrency market by being the first to quantify and analyse common volatility (COVOL) across different subsectors. Our findings highlight that the drivers of COVOL vary significantly across subsectors like Proof of Work (PoW), Proof of Stake (PoS), and Smart Contract platforms. Macroeconomic factors such as the US Dollar Index, term spreads, and interest rates have distinct impacts on these subsectors. This suggests that investors should consider both the economic environment and the specific characteristics of each subsector when managing their portfolios. Policymakers, in turn, need to account for these variations when developing regulations aimed at addressing systemic risks and maintaining market stability.

Furthermore, the study uncovers unique volatility profiles across these subsectors. Major cryptocurrencies like Bitcoin and Ethereum tend to exhibit more stability, while newer cryptocurrencies that form the PoS and Smart Contract platforms subsectors, are more sensitive to market-wide shocks. This has significant implications for both investors and regulators. Investors need to differentiate their strategies based on the sensitivity of individual cryptocurrencies within these subsectors, recognising that highly sensitive assets may pose greater risks during periods of high volatility. Policymakers can use these insights to prioritise regulation that focuses on stabilising more volatile subsectors to prevent market disruptions.

In summary, this paper underscores the importance of subsector-specific COVOL analysis in understanding cryptocurrency market dynamics. By identifying which subsectors are more vulnerable to macroeconomic shocks and different financial risks, the study provides a more nuanced understanding of risk in the cryptocurrency space. This analysis is crucial for both investors seeking

tailored risk management strategies and policymakers aiming to develop regulatory frameworks that address the distinct risks posed by different subsectors in the rapidly evolving cryptocurrency market.

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Table 1. Descriptive statistics*Panel A. Proof of Work (PoW) subsector*

Ticker	N. Obs.	Mean	SD	Skewness	Kurtosis	JB	ADF	Q	Q2
BTC	2956	0.07	1.65	-0.84	16.32	22213.37***	-13.14***	16.47**	80.79***
ETH	2370	0.09	2.45	-0.58	11.06	6543.52***	-11.84***	21.37***	135.72***
DOGE	2506	0.06	3.20	4.71	97.04	932643.62***	-12.56***	44.79***	50.98***
BCH	1967	0.02	2.47	0.05	20.06	23862.92***	-12.53***	26.49***	57.5***
LTC	2789	0.04	2.43	0.64	17.04	23108.24***	-13.25***	28.77***	176.04***
ETC	2816	0.04	2.69	0.23	12.69	11046.23***	-13.95***	24.76***	231.11***

Panel B. Proof of Stake (PoS) subsector

Ticker	N. Obs.	Mean	SD	Skewness	Kurtosis	JB	ADF	Q	Q2
ETH	587	0.04	1.31	-0.40	9.36	1003.84***	-8.23***	16.09**	166.39***
SOL	1367	0.15	3.10	-0.03	10.42	3134.02***	-9.83***	26.35***	179.95***
ADA	1755	0.04	2.29	-0.36	12.05	6026.65***	-10.56***	39.28***	110.28***
AVAX	1203	0.08	2.91	0.43	10.94	3195.08***	-9.55***	24.1***	114.72***
NEAR	1283	0.06	3.03	0.05	7.38	1026.5***	-10.63***	12.3**	54.02***
APT	550	0.01	2.54	0.90	12.34	2072.98***	-6.71***	22.56***	57.31***
MATIC	1755	0.09	3.14	-0.31	18.56	17730.04***	-11.41***	45.46***	102.77***
FIL	1093	-0.13	2.53	-0.26	10.60	2639.6***	-9.71***	26.98***	58.02***

Panel C. Smart contract platform subsector

Ticker	N. Obs.	Mean	SD	Skewness	Kurtosis	JB	ADF	Q	Q2
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BNB	2134	0.07	2.12	-0.42	24.51	41208.04***	-11.17***	52.31***	223.3***
TRX	2134	0.02	2.12	-0.86	17.16	18092.05***	-12.75***	36.49***	115.91***
LINK	1765	0.04	2.57	-0.86	14.36	9701.55***	-12.35***	29.9***	98.1***
ICP	1073	-0.13	2.65	0.23	8.31	1268.35***	-10.25***	13.15*	124.42***

Note: This table reports the descriptive statistics of excess return series of 17 cryptocurrencies in the sample between 11 March 2016 and 23 April 2024. ETH is classified as a PoW cryptocurrency from 11 March 2016 to 14 September 2022. Since 15 September 2022, ETH is considered PoS cryptocurrency. LB-Q(10) and LB-Q(20) represent the Ljung-Box Q-statistics up to the 10th and 20th order autocorrelation. Jarque-Bera statistics indicate the test for the normality of sample data. ADF test represent the Augmented Dickey-Fuller (ADF) unit root test. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

Table 2. The largest estimated common volatility for PoW subsector*Panel A. Twenty largest COVOL*

Date	$x_{PoW,t}^g$	$\bar{r}_{PoW}$	r_{SP500}	r_{WTI}	r_{Gold}	Event
28/01/2021	152.72	14.47	0.97	-0.97	-0.20	Cryptocurrency market bubble in 2021
02/03/2024	51.42	6.05	NA	NA	NA	Record fund inflows to Bitcoin ETFs
30/03/2017	42.51	8.52	0.30	1.68	-0.73	Cryptocurrency market bubble in 2017
17/06/2016	40.37	-6.63	-0.33	NA	3.83	Bitcoin's sell-off about 25 percent in 6 days
12/07/2017	34.74	-2.45	-0.1	-0.66	0.04	Sell-off across major cryptocurrencies including BTC and ETH
11/10/2018	33.23	1.98	-0.44	1.06	-0.56	NYSE announced Bakkt, a federally regulated digital asset exchange
29/10/2022	25.96	3.97	NA	NA	NA	Cryptocurrency market bubble burst in 2022: FTX bankruptcy
12/03/2020	25.85	-22.18	-10.00	-4.59	-3.61	COVID-19 global sell-offs
31/08/2018	24.71	3.42	0.01	-0.64	0.10	NYSE announced Bakkt, a federally regulated digital asset exchange
01/09/2017	23.20	6.12	0.20	0.13	0.22	LTC price was up by 26.1%
23/12/2016	23.17	1.76	0.12	0.13	0.43	BTC's price surged by 15% from USD 800 to more than USD 900 in one week
02/01/2021	21.72	6.51	NA	NA	NA	Crypto market cap surged above USD 1 trillion for the first time
12/06/2018	21.25	-0.88	0.17	0.39	-0.32	Cryptocurrency market bubble burst in 2018
12/10/2017	19.60	2.74	-0.17	-1.36	0.14	BTC's price soared above USD 5,000 to record high
02/04/2019	19.26	8.37	0.00	1.59	0.43	Cryptocurrency market rally after 2018 sell-off: BTC topped USD 5k for the first time in 2019
22/07/2016	18.90	-1.94	0.45	-1.25	-0.65	Post Dao-hack event; Bitfinex went offline intraday
05/09/2022	18.66	1.83	-0.20	0.80	-0.07	Russia and Iran accepted crypto for cross-border payments
04/01/2023	18.19	2.19	0.75	-5.35	0.79	Broad cryptocurrency market rally
08/07/2020	18.02	4.38	0.78	0.69	0.86	US Regulator (OCC) green-lighted banks for Cryptocurrency Custody Services

16/04/2021	17.71	-2.98	1.10	-0.52	0.74	Cryptocurrencies tumbled amid Turkey's central bank ban of crypto payments
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Panel B. Estimated factor loadings of each cryptocurrency

Cryptocurrency	Factor loading
DOGE	0.4587
ETC	0.4287
BCH	0.3994
LTC	0.3969
ETH	0.3826
BTC	0.3773

Note: Panel A of Table 2 presents the dates with the largest common volatility of the PoW cryptocurrencies. x_t^σ denotes the common volatility. r_{SP500} , r_{WTI} , and r_{Gold} denotes the returns on the S&P 500 index, the WTI crude oil futures, and the gold futures. $\bar{r}_{PoW}$ indicates the cross-sectional mean return of PoW cryptocurrencies. Panel B lists the cryptocurrencies and their factor loadings, estimated from the empirical model described in subsection 3.1.

Table 3. The largest estimated common volatility for PoS subsector*Panel A. Twenty largest COVOL*

Date	$x_{PoS,t}^G$	$\bar{r}_{PoS}$	r_{SP500}	r_{WTI}	r_{Gold}	Event
10/12/2019	45.43	-15.40	-0.11	0.37	0.15	Houbi's US-based crypto trading platform ceased operation.
15/10/2021	29.13	1.96	0.74	1.19	-1.59	Trustology received full FCA registration as a UK registered cryptoassets business
30/07/2022	27.97	-2.37	NA	NA	NA	Cryptocurrency market bubble burst in 2022
06/08/2019	27.48	4.64	1.30	-1.96	0.71	U.S. House of Representatives presented a new bill to stop double taxation on crypto
26/05/2020	25.51	4.00	1.22	1.26	-0.99	COVID-19 pandemic
01/07/2020	23.95	3.02	0.50	1.39	-0.64	COVID-19 pandemic
24/09/2019	22.09	-6.96	-0.85	-2.33	0.67	Broad cryptocurrency market sell-off
25/11/2019	21.27	7.77	0.75	0.41	-0.50	Broad cryptocurrency market rebound
07/09/2021	20.16	1.63	-0.34	-0.66	-1.60	El Salvador accepted Bitcoin as legal tender fuelling the crypto bubble in 2021
17/08/2021	19.97	0.94	-0.71	-1.05	-0.07	Ethereum's London Hard Fork EP-1559 upgrade went live
17/02/2023	18.85	3.64	-0.28	-2.78	0.23	Hot inflation data raised fears in financial markets
08/10/2019	18.15	2.69	-1.57	-0.23	0.79	Binance launched P2P Trading for Chinese Yuan
09/01/2023	16.88	3.22	-0.08	1.18	0.31	Price of Solana (SOL) increased by a massive of 45% in the past week
25/07/2020	16.57	3.84	NA	NA	NA	COVID-19 pandemic: Cardano surged 67.5% in July 2020
16/10/2020	16.36	-4.99	0.01	-0.20	-0.46	Filecoin, designed as a decentralized alternative to Amazon Web Services, went live.
26/02/2020	15.78	-3.82	-0.38	-2.37	0.28	COVID-19 global sell-offs
01/03/2022	15.32	1.24	-1.55	8.03	1.85	Heightened geopolitical risk due to Russia-Ukraine war

16/12/2022	13.34	-6.33	-1.11	-2.36	0.88	Post FTX's bankruptcy
05/09/2020	13.21	-7.54	NA	NA	NA	COVID-19 period: Cryptocurrency market correction
29/06/2019	12.70	5.25	NA	NA	NA	Broad cryptocurrency market rebound

Panel B. Estimated factor loadings of each cryptocurrency

Cryptocurrency	Factor loading
ADA	0.3972
APT	0.3892
MATIC	0.3791
FIL	0.3740
SOL	0.3430
NEAR	0.3330
AVAX	0.3252
ETH	0.2698

Note: Panel A of Table 3 presents the dates with the largest common volatility of the PoS cryptocurrencies. x_t^σ denotes the common volatility. r_{SP500} , r_{WTI} , and r_{Gold} denotes the returns on the S&P 500 index, the WTI crude oil futures, and the gold futures. $\bar{r}_{PoS}$ indicates the cross-sectional mean return of PoS cryptocurrencies Panel B lists the cryptocurrencies and their factor loadings, estimated from the empirical model described in subsection 3.1.

Table 4. The largest estimated common volatility for Smart Contract Platform subsector*Panel A. Twenty largest COVOL*

Date	$x_{Smart,t}^G$	$\bar{r}_{Smart}$	r_{SP500}	r_{WTI}	r_{Gold}	Event
31/03/2021	49.26	5.26	0.36	-2.32	1.31	Cryptocurrency market bubble in 2021
21/04/2022	29.03	2.28	-1.48	1.01	-0.30	Cryptocurrency market bubble burst in 2022
03/09/2020	27.23	-3.69	-3.57	-0.34	-0.63	Cryptocurrency market correction
24/09/2019	24.45	-7.11	-0.85	-2.33	0.67	Cryptocurrency sell-off: BTC deepened under USD 10k
28/11/2018	21.09	-2.80	2.27	-2.49	0.53	BTC's market cap fell below USD 100 billion for the first time since October 2017 and its price declined to USD 5.5k.
21/10/2023	20.21	2.30	NA	NA	NA	Cryptocurrency market bubble in 2023
18/05/2021	17.73	-0.21	-0.86	-1.18	0.11	Ethereum crossed USD 3,000 for the first time, worth more than BoA and Walt Disney.
24/03/2019	17.39	1.72	NA	NA	NA	Many large financial institutions entered the crypto custody services
21/12/2018	16.59	2.45	-2.08	-0.63	-0.33	G20 decided on regulating the crypto sector
20/07/2023	15.74	2.15	-0.68	0.37	-0.36	Broad cryptocurrency market rally
25/07/2021	15.37	2.40	NA	NA	NA	Wyoming was first US state to legally recognize DAO
19/02/2021	15.31	10.14	-0.19	-2.14	0.38	Microstrategy topped up bitcoin purchase, becoming one of the largest institutional holders of bitcoin
21/09/2018	13.99	4.32	-0.04	-0.03	-0.67	Cryptocurrency market rebound from the Great Crypto Crash
02/06/2022	13.41	3.18	1.83	1.39	1.22	Cryptocurrency market bubble burst in 2022
26/05/2019	13.37	1.18	NA	0.24	NA	BNB rallied fivefold in H1/2019
05/02/2019	13.36	1.52	0.47	-1.66	0.25	BNB rallied fivefold in H1/2019
26/02/2020	13.05	-2.94	-0.38	-2.37	0.28	COVID-19 global sell-offs

08/08/2020	12.83	4.10	NA	NA	NA	Cryptocurrency market cap recorded a new yearly high at USD 343 billion.
04/05/2021	11.95	-0.90	-0.67	1.84	-0.78	Ethereum crossed USD 3,000 for the first time, worth more than BoA and Walt Disney.
27/06/2021	11.75	5.43	NA	NA	NA	Broad cryptocurrency market rebound

Panel B. Estimated factor loadings of each cryptocurrency

Cryptocurrency	Factor loading
TRX	0.5136
ICP	0.5056
BNB	0.4918
LINK	0.4885

Note: Panel A of Table 4 presents the dates with the largest common volatility of the Smart Contract Platform cryptocurrencies. x_t^{σ} denotes the common volatility. r_{SP500} , r_{WTI} , and r_{Gold} denotes the returns on the S&P 500 index, the WTI crude oil futures, and the gold futures. $\bar{r}_{Smart}$ indicates the cross-sectional mean return of smart contract platform cryptocurrencies Panel B lists the cryptocurrencies and their factor loadings, estimated from the empirical model described in subsection 3.1.

Table 5. Validation tests*Panel A. Using cryptocurrency market volatility as dependent variable (ϑ_m^{BMD})*

	PoW	PoS	Smart
	(1)	(2)	(3)
$COVOL_{subsector}^2$	0.31*** (0.09)	0.15** (0.08)	0.09** (0.04)
Δ_m^{GEPU}	0.01 (0.01)	0.01 (0.02)	0.02 (0.03)
Δ_m^{VIX}	0.17** (0.08)	0.21** (0.13)	0.11 (0.09)
Δ_m^{GOPRX}	0.03 (0.03)	0.02 (0.04)	0.03 (0.06)
Observations	98	59	71
R ²	0.1537	0.1465	0.1042
Adjusted R ²	0.1142	0.1274	0.0698
F Statistic	6.72***	5.91***	3.67**

Panel B. Using average correlation of cryptocurrencies in the sample as dependent ($\rho_{subsector,m}$)

	PoW	PoS	Smart
	(1)	(2)	(3)
$COVOL_{subsector}^2$	0.02** (0.008)	0.014*** (0.004)	0.012** (0.005)
Δ_m^{GEPU}	0.0002 (0.0002)	0.0004 (0.0003)	0.0003 (0.0002)
Δ_m^{VIX}	0.004 (0.003)	0.006 (0.006)	0.005 (0.004)
Δ_m^{GOPRX}	-0.002 (0.003)	-0.001 (0.003)	-0.001 (0.001)
$\rho_{subsector,m-1}$	0.45*** (0.11)	0.37*** (0.09)	0.31*** (0.09)
Observations	98	59	71
R ²	0.2423	0.2661	0.2468
Adjusted R ²	0.1934	0.2198	0.2208
F Statistic	5.59***	5.68***	6.25***

Note: Table 5 presents the regression results (estimated coefficients and their standard errors) of validation tests, with Panels A and B using the cryptocurrency market's volatility (ϑ_m^{BMD}) and subsector

mean correlation ($\rho_{subsector,m}$) as dependent variable, respectively. $COVOL_{subsector}^2$ is the squared monthly common volatility of each subsector cryptocurrency market including PoW, PoS, and Smart Contract; Δ_m^{GEPU} denotes the monthly log-differenced Global Economic Policy Uncertainty Index (GEPU); Δ_m^{VIX} indicates the monthly log-differenced of the CBOE Volatility Index (VIX); Δ_m^{GOPRX} represents the monthly log-differenced of the Global Geopolitical Risk Index (GOPRX). Newey and West's (1987) robust standard errors are used. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

Table 6. Determinants of subsectors' COVOL – Full sample

	PoW	PoS	Smart
	(1)	(2)	(3)
<i>RAI</i>	-0.005 (0.01)	0.32 (0.24)	-0.02** (0.01)
<i>SP500</i>	0.0002 (0.03)	-0.06 (0.19)	0.02 (0.02)
<i>DXY</i>	-0.002 (0.003)	-0.005 (0.05)	-0.002 (0.004)
<i>Term10Y2Y</i>	0.12** (0.06)	0.72** (0.36)	0.01 (0.04)
<i>DGS2</i>	-0.03 (0.02)	0.29** (0.13)	0.01 (0.02)
Intercept	1.18*** (0.35)	3.42 (3.74)	0.99*** (0.36)
N. Obs.	2,959	1,755	2,135
Adj. R-Squared	0.0041	0.0032	0.0028
F-statistics	2.52*	2.01*	1.78

Note: This table presents the regression results of Eq. (8) to investigate the daily determinants of the COVOL of the cryptocurrency market and subsectors for the whole research period. Eq. (8) is estimated using OLS estimation with t-statistics computed using Newey and West's (1987) robust standard errors. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

Table 7. Determinants of subsectors' COVOL – Monthly Data

	PoW	PoS	Smart
	(1)	(2)	(3)
<i>RAI</i>	-0.02 (0.03)	0.11 (0.07)	-0.004 (0.01)
<i>SP500</i>	-0.04 (0.10)	-0.13 (0.94)	-0.12* (0.06)
<i>DXY</i>	0.004 (0.005)	0.005 (0.06)	0.002 (0.004)
<i>Term10Y2Y</i>	0.11* (0.06)	0.55** (0.21)	0.09** (0.04)
<i>DGS2</i>	0.03 (0.02)	0.13 (0.78)	0.05*** (0.01)
<i>MPU</i>	0.02 (0.04)	0.91** (0.41)	-0.02 (0.02)
<i>UCRY_{Policy}</i>	-0.012** (0.005)	-0.03 (0.02)	-0.01*** (0.003)
<i>UCRY_{Price}</i>	0.002** (0.001)	0.021*** (0.007)	0.004** (0.002)
Intercept	1.73*** (0.32)	2.12 (1.83)	1.84*** (0.27)
N. Obs.	98	59	71
Adj. R-Squared	0.1452	0.1214	0.1611
F-statistics	2.22**	1.98**	2.44**

Note: This table presents the regression results of the augmented version of Eq. (8) to investigate the monthly determinants of the COVOL of the cryptocurrency market and subsectors for the whole research period. The model is estimated using OLS estimation with t-statistics computed using Newey and West's (1987) robust standard errors. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.

ONLINE APPENDICES

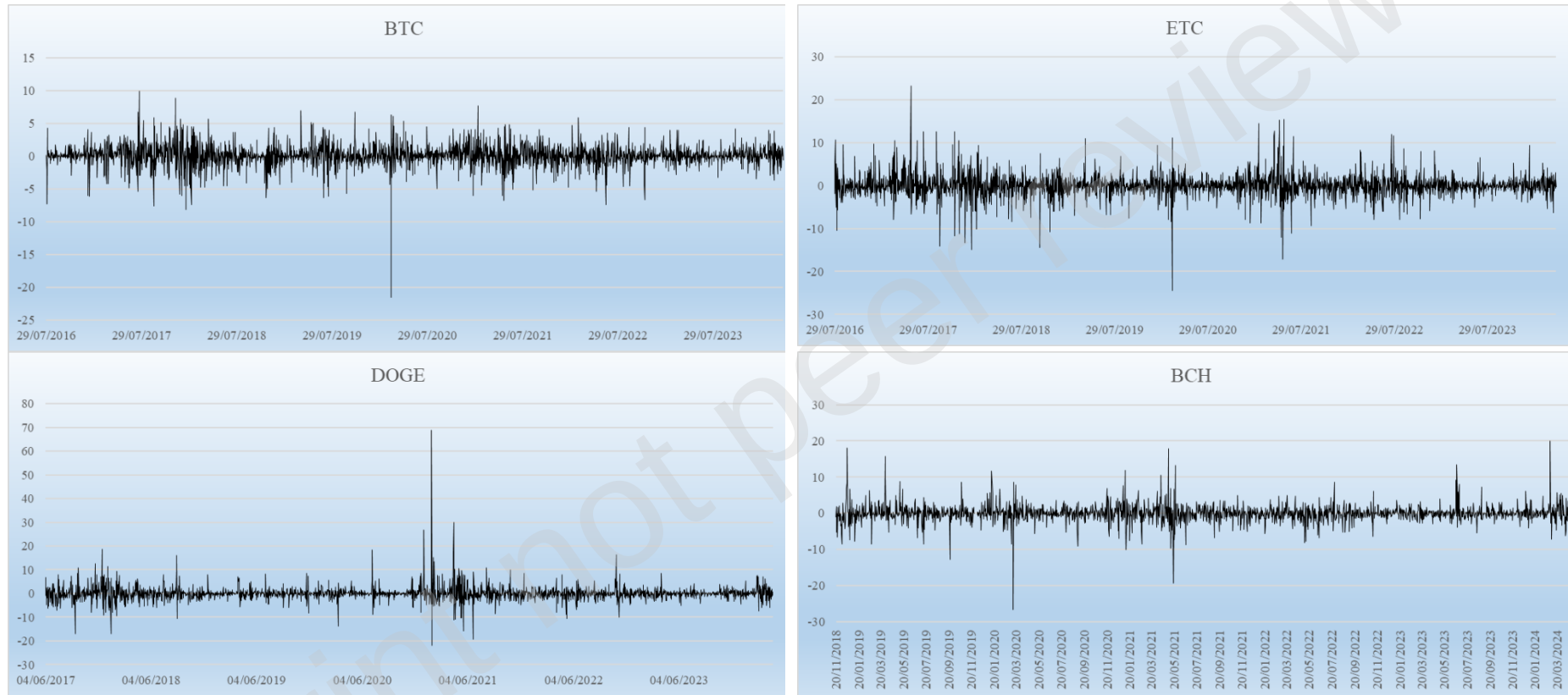
Online Appendix A1. Cryptocurrency description and market capitalisation

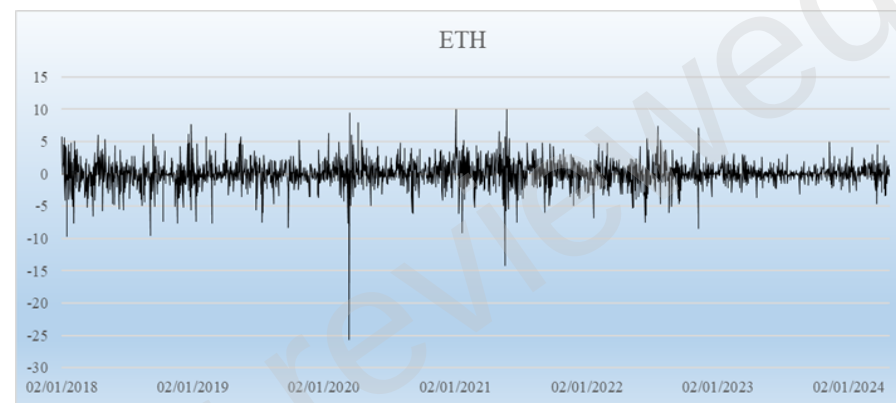
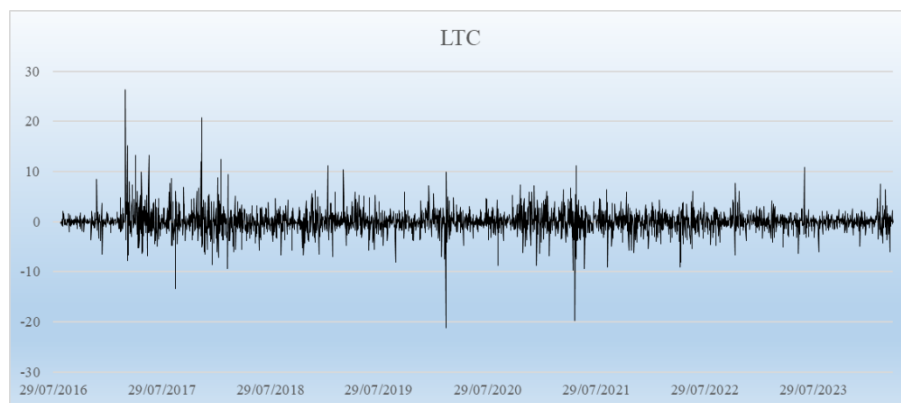
No.	Ticker	Full Name	Subsector	Description	Market Cap (billions of USD)
1	BTC	Bitcoin	Proof of Work (PoW)	Decentralized digital currency, first and largest cryptocurrency	\$1200
2	ETH	Ethereum	Proof of Stake (PoS) after September 2022; Proof of Work (PoW) before September 2022	Smart contract platform and cryptocurrency	\$600
3	BNB	Binance Coin	Smart Contract	Utility token of the Binance exchange	\$60
4	SOL	Solana	Proof of Stake (PoS)	High-performance blockchain platform	\$25
5	DOGE	Dogecoin	Proof of Work (PoW)	Meme cryptocurrency, originally created as a joke	\$28
6	ADA	Cardano	Proof of Stake (PoS)	Blockchain platform with Proof of Stake consensus	\$43
7	AVAX	Avalanche	Proof of Stake (PoS)	Scalable blockchain platform with consensus protocol	\$10
8	BCH	Bitcoin Cash	Proof of Work (PoW)	Fork of Bitcoin with larger block sizes	\$9
9	TRX	Tron	Smart Contract	Blockchain-based decentralized content platform	\$20
10	LINK	Chainlink	Smart Contract	Oracle network connecting smart contracts with real-world data	\$8
11	NEAR	NEAR Protocol	Proof of Stake (PoS)	Scalable blockchain platform	\$7.6
12	MATIC	Polygon	Proof of Stake (PoS)	Ethereum layer-2 scaling solution	\$15
13	ICP	Internet Computer	Smart Contract	Blockchain platform for decentralized applications	\$5.5
14	LTC	Litecoin	Proof of Work (PoW)	Peer-to-peer cryptocurrency	\$14
15	APT	Aptos	Smart Contract	Layer-1 blockchain platform	\$3.6
16	ETC	Ethereum Classic	Proof of Work (PoW)	Ethereum fork focused on immutability	\$4
17	FIL	Filecoin	Proof of Stake (PoS)	Decentralized storage network	\$3.1

Note: This table presents the name, ticker, description, and market capitalisation as of 10 April 2024 of the 25 cryptocurrencies in our sample. The market capitalisation data is sourced from coingecko.com.

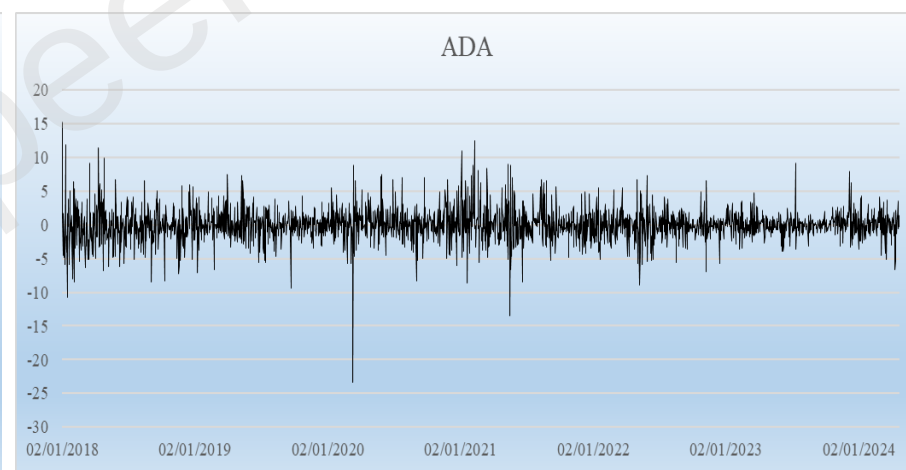
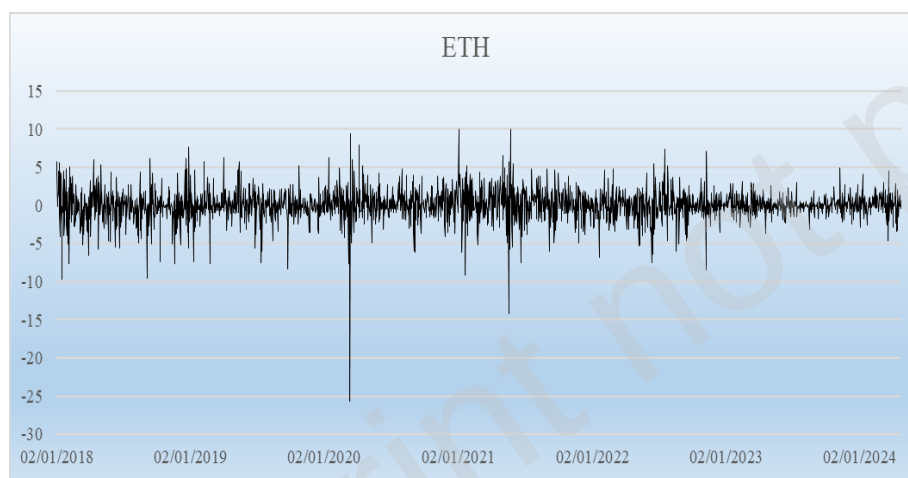
Online Appendix A2. Cryptocurrency daily return series

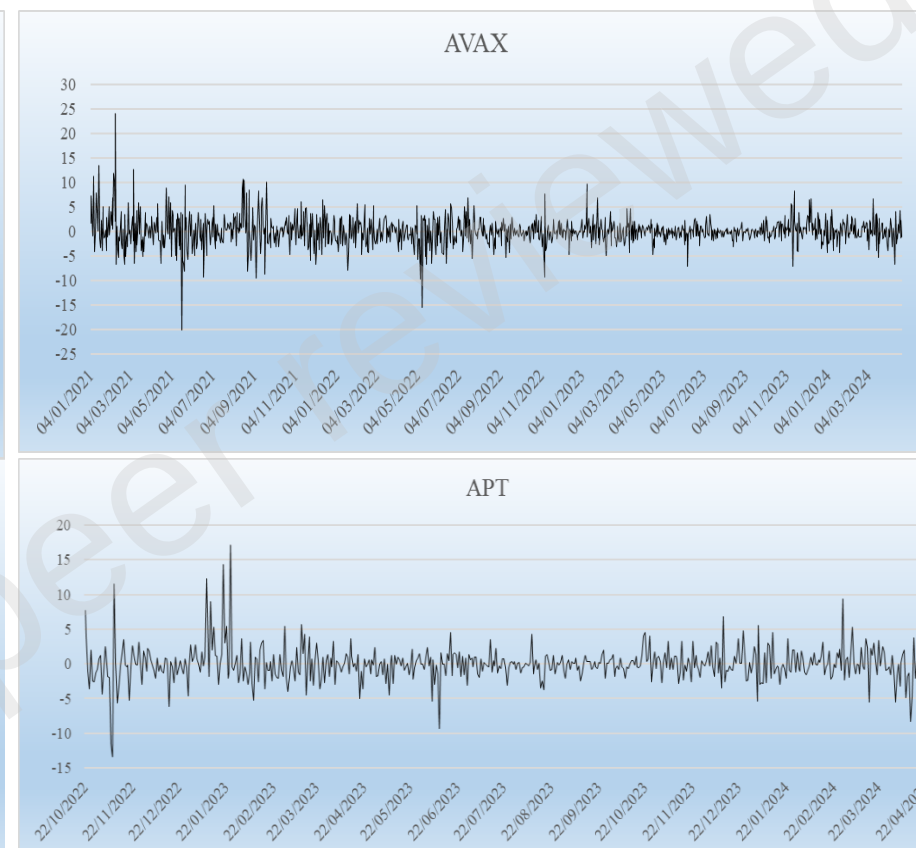
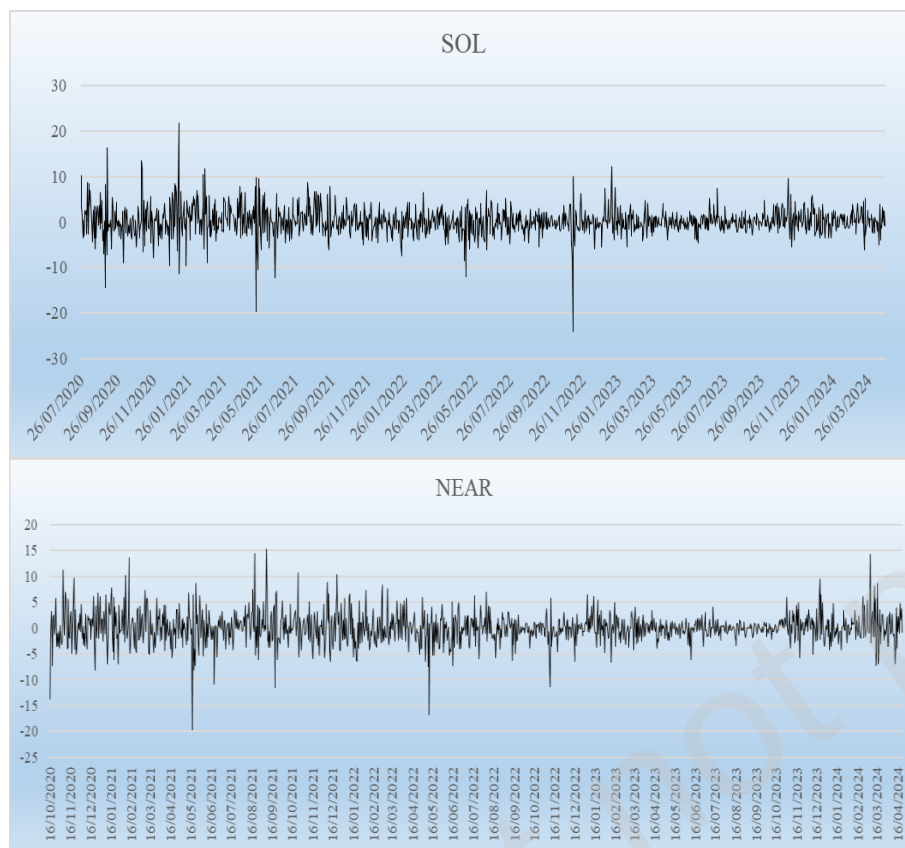
Panel A21. Proof of Work

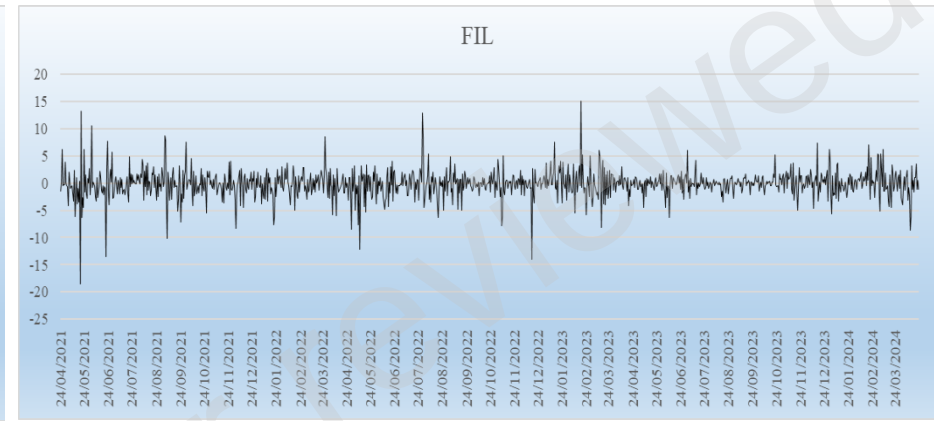
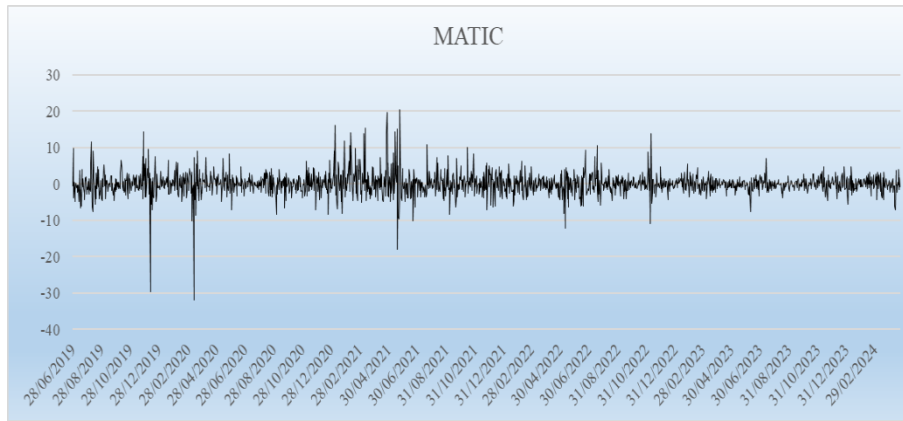




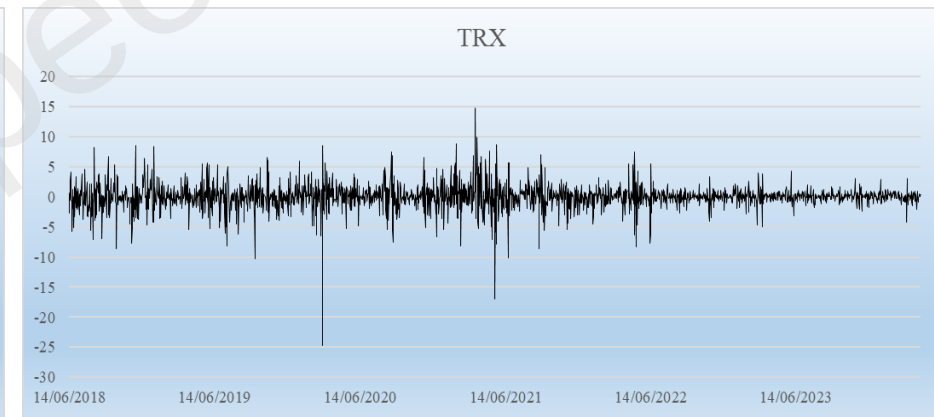
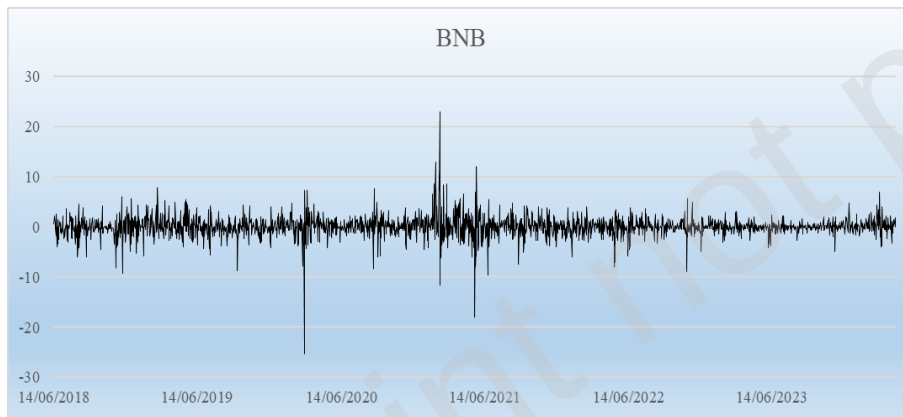
Panel A22. Proof of Stake

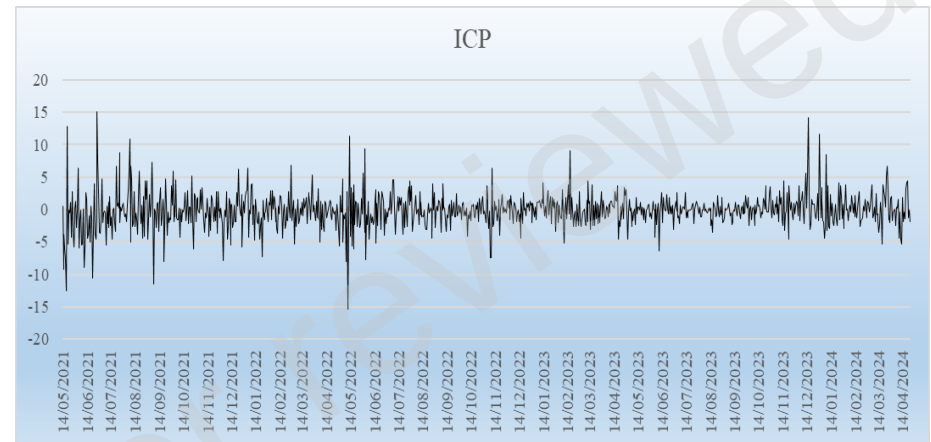
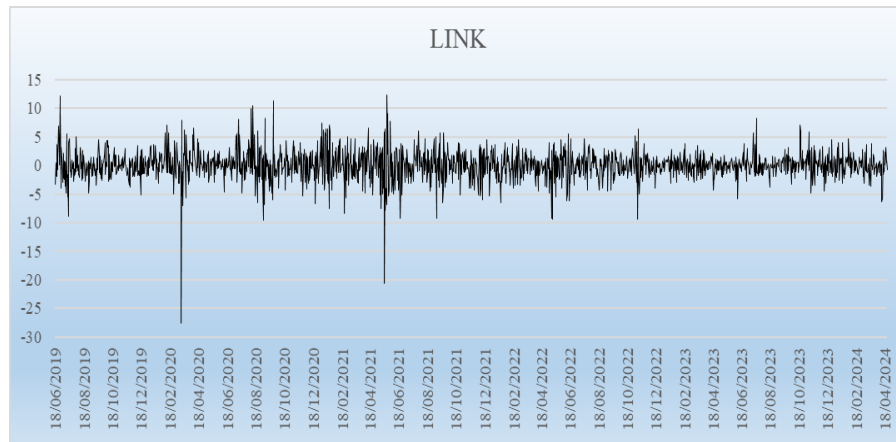






Panel A23. Smart Contract Platform





Online Appendix 3. Dynamic COVOL

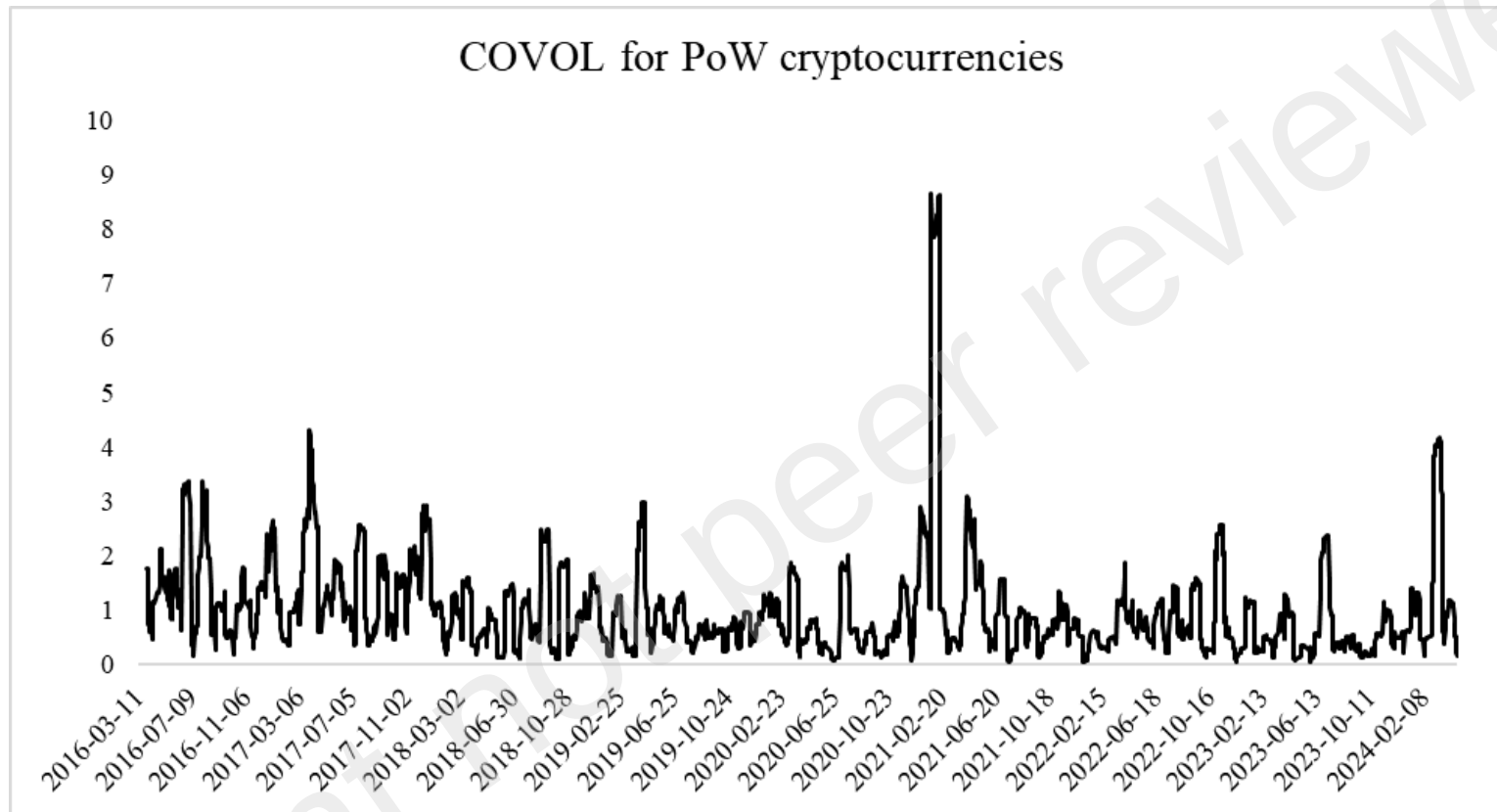


Figure 1. The 20-day moving average of the COVOL for PoW cryptocurrencies

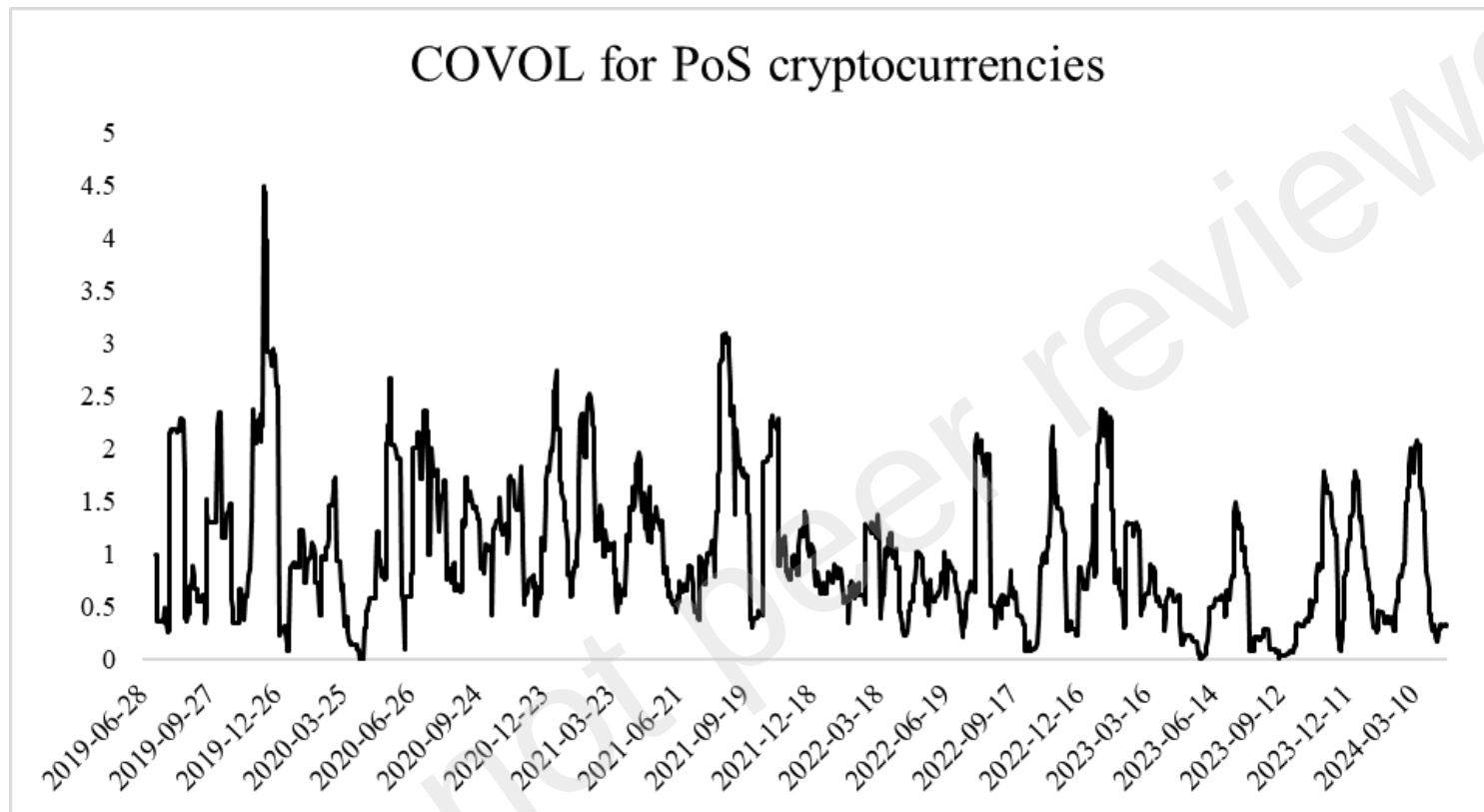


Figure 2. The 20-day moving average of the COVOL for PoS cryptocurrencies

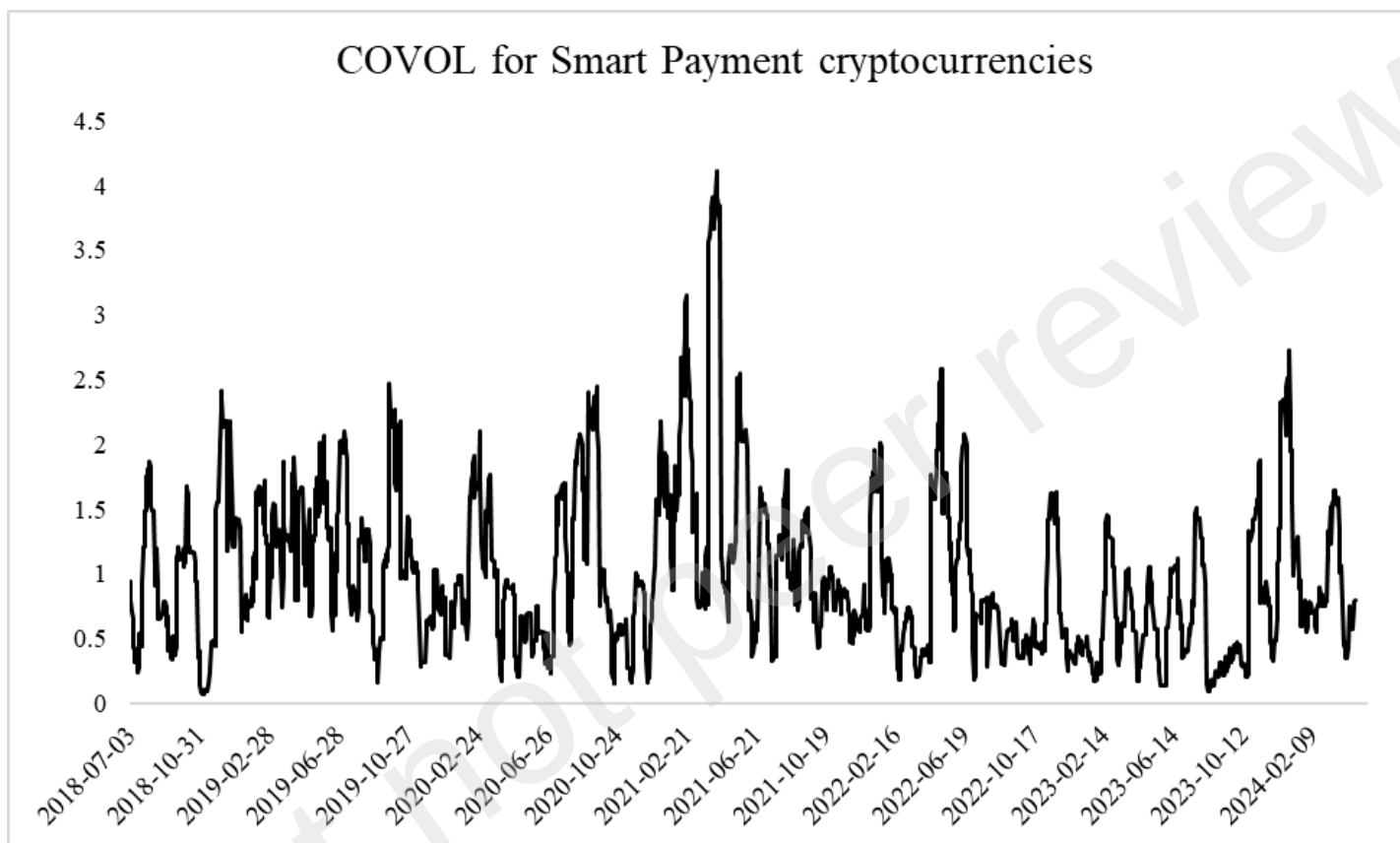


Figure 3. The 20-day moving average of the COVOL for Smart Contract Platform cryptocurrencies

Online Appendix 4. Determinants of subsectors' COVOL – Crisis periods

Panel A. COVID-19 period

	PoW	PoS	Smart
	(1)	(2)	(3)
<i>RAI</i>	0.35** (0.17)	-0.016** (0.008)	-0.02* (0.01)
<i>SP500</i>	-0.20 (0.13)	-0.01 (0.01)	0.03 (0.04)
<i>DXY</i>	-0.19** (0.08)	0.02** (0.01)	0.004 (0.01)
<i>Term10Y2Y</i>	-1.56 (1.87)	0.17 (0.31)	0.17 (0.26)
<i>DGS2</i>	0.44 (0.35)	0.08 (0.11)	0.11 (0.09)
Intercept	16.11* (8.76)	-2.29 (1.65)	0.38 (1.19)
N. Obs.	225	225	225
Adj. R-Squared	0.1151	0.0782	0.033
F-statistics	5.98***	4.32**	2.49*

Panel B. Russia-Ukraine war period

	PoW	PoS	Smart
	(1)	(2)	(3)
<i>RAI</i>	-0.12 (-0.37)	0.14 (0.21)	-0.11** (0.06)
<i>SP500</i>	-0.21 (0.15)	0.03 (0.04)	0.04 (0.04)
<i>DXY</i>	0.02 (0.04)	-0.002 (0.01)	0.003 (0.01)
<i>Term10Y2Y</i>	0.98* (0.52)	0.49*** (0.17)	0.41*** (0.11)
<i>DGS2</i>	0.13 (0.12)	-0.10 (0.12)	0.18** (0.06)
Intercept	0.54 (2.71)	0.87 (0.89)	0.35 (0.89)
N. Obs.	206	206	206

Adj. R-Squared	0.0212	0.086	0.045
F-statistics	1.98*	3.91***	1.89*

Note: This table presents the regression results of Eq. (8) to investigate the daily determinants of the COVOL of the cryptocurrency market and subsectors for the crisis periods. The period for the COVID-19 pandemic is between January 30, 2020, to December 31, 2020. The period for the Russia-Ukraine war is between April 22, 2022, to December 31, 2022. Eq. (8) is estimated using OLS estimation with t-statistics computed using Newey and West's (1987) robust standard errors. *, **, and *** indicate statistical significance at 10%, 5%, and 1% level, respectively.